A Binary-Schedule Benchmark and Diagnostic Study for Quantum-Ready Robust Low-Thrust Cislunar Maneuver Initialization

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Abstract

Low-thrust cislunar maneuver design couples nonlinear continuous trajectory optimization with discrete choices about thrust, coast, and recovery timing. This paper studies a quantum-ready binary-schedule initialization workflow in which a thrust-window objective with missed-thrust penalties is approximated by a QUBO surrogate, sampled by classical heuristics and statevector QAOA, and refined by norm-bounded classical branch-recovery optimization. The study is framed as a reproducible benchmark and diagnostic package, not as a replacement for deterministic optimal control or as evidence of quantum advantage. The evidence is deliberately stratified. The unrepaired hard catalog-derived and phase-shift binary samplers often fail or collapse to dense all-windows schedules, while a duty-cycle-prior $N = 12$ benchmark and a 30-seed main-method package show that several classical methods and surrogate-QUBO simulated annealing can find refinable high-duty one-coast schedules. The configured statevector QAOA row is not competitive with the strongest classical methods in the main 30-seed package, and a separate optimized-angle QAOA-depth ablation does not establish superiority over surrogate-QUBO simulated annealing. Continuous-backend diagnostics give more positive but separate evidence: bounded multiple shooting, continuation, compact direct collocation, and Hermite–Simpson probes solve limited non-teacher phase-shift rows, while the selected-outage hard-catalog envelope remains negative because nominal thresholds fail and all-mask diagnostics remain high. For the hard catalog-DRO fixed-final-time case, the strongest tail-coast locked-nominal evidence is now a combined configured one/two-segment row: with the final five nominal controls fixed exactly to zero, `tail_coast_all_one_two_segment_t5_portfolio` selects and evaluates all 27 configured masks with `outage_lengths=[1,2]`, reaches nominal tail-coast error 0.022992 and selected/all fixed-time worst error 0.093606 under the 0.09/0.17 thresholds, and sets `meets_thresholds=true`. Branches with post-outage recovery variables are optimized, while two late-tail no-recovery-variable masks are threshold-feasible direct evaluations rather than optimizer-converged branches, so `branch_optimizer_all_success=false`. Deterministic fallback starts are declared and charged, with four fallback-evaluated branches accepted by fallback starts. Separate all-one-segment and all-two-segment rows are retained as provenance/scope rows. These tail-coast results are not QUBO, QAOA, quantum, fuel-optimality, high-fidelity mission-design, or production optimal-control evidence. The remaining bottlenecks are dense continuous-control initialization, recovery parameterization beyond the configured outage family, surrogate generalization, and statistically significant improvement over strong classical initializers under equal total trajectory-evaluation budgets.

Keywords: Low-thrust trajectory optimization, quantum annealing, QAOA, QUBO, cislunar dynamics, missed-thrust events, CR3BP

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1. Introduction

Electric low-thrust propulsion enables propellant-efficient cislunar and interplanetary transfers, but it also turns maneuver planning into a long-horizon nonlinear optimal-control problem. A practical low-thrust design must choose continuous thrust magnitudes and directions while also choosing a qualitative maneuver structure: which time intervals should thrust, coast, or recover from an outage. The continuous part is usually handled by direct transcription, collocation, multiple shooting, sequential convex programming, or indirect methods. The discrete structure is rarely searched exhaustively, yet it strongly affects whether a deterministic optimizer converges.

This initialization sensitivity is especially important for robust low-thrust design. Missed-thrust events caused by safe modes, propulsion interruptions, or operational constraints can invalidate an otherwise efficient transfer. Grebow and McCarty show that including missed-thrust branches in low-thrust cislunar design can greatly reduce recovery propellant for week-scale outages [1]. Sinha and Beeson identify initial-guess generation as a bottleneck for low-thrust trajectory design with missed-thrust robustness [2]. These results motivate the central question of this paper:

> Can a quantum-ready binary optimization layer generate useful robust low-thrust maneuver schedules when missed-thrust fragility is included in the binary energy model and the resulting schedules are refined by classical continuous optimization?

The question is timely because low-thrust trajectory optimization has recently been mapped into quantum-ready binary formulations. De Grossi et al. transcribed an Earth–Mars low-thrust trajectory optimization problem into a binary/QUBO form and tested D-Wave quantum and hybrid solvers, while emphasizing present hardware limits [3]. Moretta and Casasola studied quantum-assisted initialization for Earth–Moon low-thrust trajectories using binary thrust-activation patterns as seeds for deterministic solvers [4]. The present paper differs by making missed-thrust recovery part of the initialization objective and by reporting negative and controlled-feasible evidence side by side.

The paper makes four contributions. First, it formulates a robust binary thrust-window initialization problem in the Earth–Moon CR3BP with explicit missed-thrust outage masks. Second, it fits a QUBO surrogate to trajectory-evaluated robust costs and samples it with classical QUBO simulated annealing and a statevector QAOA simulator. Third, it connects each binary schedule to a norm-bounded branch-recovery least-squares refinement in which selected outage branches receive post-outage recovery controls. Fourth, it reports a reproducible evidence hierarchy rather than a single favorable benchmark: a hard catalog-derived NRHO-like to DRO-phase case, feasibility/multiple-shooting stress tests, a selected-outage hard-catalog negative envelope, locked-nominal fixed-final-time, delayed-arrival, and tail-coast fixed-final-time branch-recovery diagnostics, bounded non-teacher phase-shift diagnostics including trapezoidal direct collocation, constant-control and independent-midpoint-control Hermite–Simpson continuation probes with persisted control sidecars, direct multiple-shooting continuation warm starts, deterministic dense-schedule repair, duty-cycle-prior main-method statistics, 30-seed optimized-QAOA ablations, and a fully disclosed teacher-controlled feasible case.

The claims are intentionally narrow. The experiments do not show quantum advantage, do not solve high-fidelity mission design, and do not demonstrate universal missed-thrust robustness. They test whether a quantum-ready discrete layer can be integrated into a robust low-thrust initialization pipeline, whether the continuous backend can solve any non-teacher cislunar case, and where the present pipeline fails.

2. Related Work

2.1. Low-thrust trajectory optimization

Low-thrust trajectory optimization has a mature literature spanning indirect methods, direct transcription, collocation, global search, and hybrid approaches. Morante et al. survey low-thrust optimization as a hybrid optimal-control problem with both continuous and discrete structure [5]. Direct methods convert the problem into a large nonlinear program, but their convergence and final local optimum can be sensitive to the supplied trajectory and control initialization. Indirect methods can be highly efficient near a solution but introduce boundary-value sensitivity through costates.

For preliminary cislunar design, the Circular Restricted Three-Body Problem is widely used because it captures the nonlinear Earth–Moon dynamical environment while keeping algorithmic studies tractable. The NASA/JPL Three-Body Periodic Orbits API provides normalized periodic-orbit states, periods, Jacobi constants, stability indices, and system constants for families such as halo orbits and distant retrograde orbits [6]. The catalog-derived benchmark in this paper uses those states as transparent boundary-condition seeds.

2.2. Missed-thrust robustness

Missed-thrust design has been studied through margin analysis, expected-thrust formulations, virtual recovery trajectories, robust optimal control, and branching trajectory methods. Rubinsztejn et al. embed expected thrust fraction into deterministic trajectory design and report improved resilience to missed-thrust scenarios [7]. Venigalla et al. formulate multi-objective low-thrust trajectory optimization with missed-thrust recovery margin [8]. Grebow and McCarty specifically study low-thrust cislunar transfers, including NRHO-to-DRO transfers, and optimize branching trajectories for outage recovery [1]. Sinha and Beeson focus on initial-guess generation for robust low-thrust missed-thrust design and compare conditional and nonconditional global-search strategies [2]. Robust cislunar low-thrust optimization has also advanced through sequential convex programming and covariance steering; Kumagai and Oguri use chance-constrained covariance steering to design nominal trajectories and correction policies under uncertainty in the Earth–Moon CR3BP [9]. These works set a high bar for any initialization method: a useful discrete initializer must ultimately feed a continuous optimizer that can produce robust feasible trajectories.

2.3. Quantum-ready binary optimization

Quantum annealers and QAOA target discrete optimization models such as Ising Hamiltonians and QUBOs. A QUBO minimizes

$$E(x) = c + \sum_i a_i x_i + \sum_{i<j} b_{ij} x_i x_j, \qquad x_i \in \{0, 1\}, \tag{1}$$

which is convertible to the Ising form used in quantum annealing and many gate-model workflows [10, 11]. Farhi et al. introduced QAOA as a gate-model variational algorithm for approximate combinatorial optimization [12]. Shukla and Vedula formulated trajectory optimization as a discrete search problem and argued that quantum search can provide speedups for suitably discretized cases [13]. De Grossi et al. provide the closest low-thrust QUBO precedent [3]. This paper uses QUBO and QAOA only for the discrete initialization layer; the continuous trajectory remains a classical optimal-control problem.

2.4. Cislunar initial guesses and direct transcription

The targeted catalog benchmark in this paper is deliberately harder than the teacher-controlled case because cislunar transfers between stable or nearly stable periodic orbits are known to be sensitive to the continuous initial guess. Pritchett, Howell, and Grebow use collocation, trajectory stacking, orbit chaining, and continuation to compute low-thrust transfers in the restricted three-body problem, including NRHO-to-DRO examples [14]. Their results are important context for this paper: binary thrust-window sampling is not a substitute for the trajectory-stacking, natural-arc, continuation, and direct-transcription machinery that mature cislunar low-thrust solvers use to enter a feasible basin.

3. Problem Formulation

3.1. Forced CR3BP dynamics

The spacecraft state is

$$s = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top \tag{2}$$

in the normalized Earth–Moon rotating frame. With mass parameter μ, distances to the primary and secondary bodies are

$$r_1 = \sqrt{(x+\mu)^2 + y^2 + z^2}, \tag{3}$$

$$r_2 = \sqrt{(x-1+\mu)^2 + y^2 + z^2}. \tag{4}$$

The forced CR3BP equations are

$$\ddot{x} = 2\dot{y} + x - \frac{(1-\mu)(x+\mu)}{r_1^3} - \frac{\mu(x-1+\mu)}{r_2^3} + a_x, \tag{5}$$

$$\ddot{y} = -2\dot{x} + y - \frac{(1-\mu)y}{r_1^3} - \frac{\mu y}{r_2^3} + a_y, \tag{6}$$

$$\ddot{z} = -\frac{(1-\mu)z}{r_1^3} - \frac{\mu z}{r_2^3} + a_z. \tag{7}$$

The transfer duration T is divided into N uniform segments. A binary variable $x_i \in \{0,1\}$ gates thrust availability during segment i. During the discrete search stage, a deterministic target-seeking steering law maps the binary schedule to a propagated trajectory,

$$g(s) = K_r(r_f - r) + K_v(v_f - v), \qquad a_i(s) = x_i a_{\max} \frac{g(s)}{\|g(s)\| + \epsilon}. \tag{8}$$

This steering law is not claimed to be optimal. It creates a repeatable map from a bitstring to a trajectory so that discrete schedules can be ranked before continuous refinement. A separate oracle diagnostic, introduced only for the teacher-controlled benchmark, initializes refinement from the hidden continuous controls used to generate the target. That diagnostic is not a competing initializer; it tests whether the refinement formulation can recover the known feasible trajectory when continuous-control initialization is no longer the bottleneck.

3.2. Missed-thrust masks and robust binary cost

Let $\mathcal{M}$ denote a set of outage masks. Each mask zeros thrust in a one- or two-segment contiguous window. For a nominal schedule x, the outage-realized schedule is $x \odot m$. The scaled terminal error under mask m is

$$e_m(x) = \left\| \begin{bmatrix} (r_T(x \odot m) - r_f)/\ell_r \\ (v_T(x \odot m) - v_f)/\ell_v \end{bmatrix} \right\|_2. \tag{9}$$

The trajectory-evaluated robust binary objective is

$$J(x) = w_n e_0(x) + w_w \max_{m\in\mathcal{M}} e_m(x) + w_d \left(\max_{m\in\mathcal{M}} e_m(x) - e_0(x) \right) + w_p \frac{1}{N} \sum_{i=1}^{N} x_i + w_s \frac{1}{N-1} \sum_{i=1}^{N-1} |x_{i+1} - x_i|. \tag{10}$$

The terms penalize nominal miss distance, worst outage miss distance, missed-thrust degradation, active thrust fraction, and thrust/coast switching.

4. Quantum-Ready Initialization Method

4.1. QUBO surrogate

Because $J(x)$ depends on nonlinear propagation and outage evaluation, it is not itself a QUBO. The method therefore fits

$$\widehat{J}(x) = c + \sum_i a_i x_i + \sum_{i<j} b_{ij} x_i x_j \tag{11}$$

from trajectory-evaluated samples $\{x^{(k)}, J(x^{(k)})\}_{k=1}^{K}$ using ridge-regularized least squares:

$$\min_\theta \sum_{k=1}^{K} \left(J(x^{(k)}) - \widehat{J}(x^{(k)}; \theta) \right)^2 + \lambda \|\theta\|_2^2. \tag{12}$$

The fitted QUBO is then sampled by classical QUBO simulated annealing or converted to a diagonal cost Hamiltonian for statevector QAOA. The present experiments use QAOA only as a small simulated gate-model sampler; no hardware quantum advantage is implied.

4.2. Discrete initializers

The experiments compare random search, cross-entropy search, a genetic algorithm, true-objective simulated annealing, surrogate QUBO simulated annealing, and statevector QAOA. Random search and true-objective simulated annealing evaluate propagated schedules directly. Cross-entropy updates Bernoulli segment probabilities from elite samples. The genetic algorithm uses elitism, crossover, and mutation. The QUBO and QAOA methods use true trajectory evaluations to train $\widehat{J}$, then validate their best sampled schedules with the true objective.

Budget accounting is therefore reported in two ways: solver true evaluations exclude shared QUBO training, while total true evaluations include QUBO training. Equal total-evaluation comparisons are the relevant fair comparison.

4.3. Continuous branch-recovery refinement

The refinement stage converts a selected binary schedule into a continuous low-thrust candidate by optimizing piecewise-constant acceleration vectors only in active windows. Refined accelerations are projected to the Euclidean thrust ball $\|u_i\|_2 \leq a_{\max}$ before propagation and persistence. The branch-recovery mode optimizes a nominal control sequence plus selected-outage-specific post-outage recovery controls. Its least-squares residual includes nominal terminal error, selected outage terminal error, fuel regularization, control regularization, and inter-segment smoothness. A run is counted as successful only if both the nominal terminal error and the selected missed-thrust worst error are below the configured thresholds.

The paper also reports an all-mask diagnostic: after refinement on selected outages, every one- and two-segment outage mask is evaluated without claiming that all masks were optimized. This separates selected-branch recovery success from universal outage robustness.

5. Experimental Setup

5.1. Common dynamics and source state

All experiments use normalized Earth–Moon CR3BP dynamics with $\mu = 0.01215058560962404$. The common initial state is taken from the first row of an Earth–Moon L2 southern halo query with period in $[1.55, 1.65]$ TU from the NASA/JPL periodic-orbit API:

$$s_0 = [1.0324985738, 0, -0.1884844917, 0, -0.1249781287, 0]. \tag{13}$$

This state is NRHO-like in the sense that it lies in the L2 southern halo family, but the paper treats it as a normalized benchmark state, not a certified operational NRHO.

5.2. Hard catalog-derived benchmark

The hard benchmark uses a distant retrograde orbit seed from an Earth–Moon DRO query with period in $[2.0, 3.0]$ TU and Jacobi constant in $[2.9, 3.1]$:

$$s_{\mathrm{DRO}} = [0.8161260827, 0, 0, 0, 0.5076556931, 0]. \tag{14}$$

The fixed target is generated by ballistic propagation of this DRO seed for the transfer time. The main catalog-derived run uses $T = 2.7$ TU, $N = 12$, $a_{\max} = 0.055$, one- and two-segment outage masks, two pilot seeds, and threshold values selected before the feasibility sweep. This case is intentionally retained even though it is hard for the current pipeline, because it prevents the paper from reporting only a constructed favorable target.

5.3. Catalog halo phase-shift benchmark

To separate continuous-backend feasibility from teacher-generated feasibility, the repository also includes non-teacher phase-shift benchmarks derived from the same JPL L2 southern halo source state. The target is generated by zero-thrust CR3BP propagation of s_0 for a phase time of 0.3 TU, while the transfer is required to arrive at that target after $T = 0.5$ TU. Thus the target is catalog/dynamics-derived rather than produced by a low-thrust teacher, but the spacecraft must correct a phase mismatch. The first configuration uses $N = 8$, $a_{\max} = 0.2$, one-segment outage masks, and the same nominal and selected robust thresholds as the other experiments. A zero-thrust trajectory has normalized nominal terminal error 0.3758, so the case is not a do-nothing pass. This benchmark is easier than the NRHO-like to DRO-phase transfer and is included to test whether the present continuous refinement can solve at least one non-teacher cislunar case.

The second phase-shift configuration uses $N = 12$ and adds a duty-cycle prior to the binary objective:

$$J_\rho(x) = w_\rho \left(\frac{1}{N} \sum_{i=1}^{N} x_i - \rho \right)^2, \tag{15}$$

with $\rho = 11/12$ and $w_\rho = 12$. This prior is not an oracle trajectory correction. It encodes a mission-design preference for one coast window while retaining a dense low-thrust arc, and it is included because the unrepaired $N = 8$ objective over-rewards sparse late thrust before continuous refinement.

5.4. Feasibility and multiple-shooting stress tests

To test whether the hard catalog-derived failure is mainly a search issue or a refinement-capability issue, the repository includes a feasibility sweep over transfer time, thrust limit, segment count, and least-squares budget. It also includes direct multiple-shooting diagnostics with nominal and selected-outage branch states and controls, RK4 segment defects, linear state-node initialization, optional nominal-first homotopy, and row-level settings fingerprints for resumed mixed runs. The newest bounded-projected variant applies the acceleration-ball projection inside the residual unpacking path, so defect propagation, terminal residuals, control penalties, smoothness penalties, evaluation, and reported controls all use the same norm-bounded accelerations. A final targeted catalog feasibility attempt uses stronger multistart branch recovery at $T = 4.0$ TU, $a_{\max} = 0.3$, $N = 14$, and a 250-evaluation limit. A selected-outage hard-catalog envelope then probes whether higher thrust and one or three optimized outage branches close the robustness gap for the same target. These tests are not used to claim a best trajectory; they are used to locate the boundary of the present implementation.

5.5. Teacher-controlled feasible benchmark

The teacher-controlled benchmark generates the target by forward propagation from s_0 under a disclosed deterministic bounded low-thrust policy. The active schedule is

$$0000011110101010, \tag{16}$$

with active fraction 0.4375, transfer time $T = 4.0$ TU, $N = 16$, $a_{\max} = 0.065$, teacher maximum control norm 0.039, and teacher fuel 0.0625147492 in normalized acceleration-time units. The generated target is

$$\begin{aligned} s_f = [&0.9072480273, -0.0480603610, -0.1216659997, \\ &-0.1816901396, -0.0351710714, 0.1330325296]. \end{aligned} \tag{17}$$

The teacher reproduces its own target with zero propagation error under the same integrator, while its worst one- or two-segment missed-thrust outage without recovery is 0.6049 under the configured error norm. This benchmark is therefore feasible by construction but not trivial under outages. It is included to test whether the initialization/refinement architecture can recover a known feasible low-thrust transfer without hiding the construction. The oracle diagnostic initializes the refinement directly with the teacher continuous controls and is labeled separately from the normal method comparison.

5.6. Metrics

The primary metrics are refined nominal terminal error, refined selected missed-thrust worst error, all-mask diagnostic worst error, schedule active fraction, nominal and recovery fuel, total true trajectory evaluations, and refinement success rate. When schedule repair is enabled, the tables separately report the solver's best schedule, the schedule selected for refinement, the repaired refined schedule, and their active fractions, so that a sparse candidate cannot be confused with a dense refined envelope. The $N = 8$ phase-shift experiments use five deterministic seeds. The $N = 12$ cardinality-prior experiment uses ten deterministic seeds, 96 shared QUBO-training evaluations, and 24 post-QUBO candidate evaluations for QUBO/QAOA methods; equal-budget classical methods receive 120 true objective evaluations, matching the QUBO methods after shared training is charged. A 30-seed main-method statistics package repeats all seven configured methods over 30 deterministic seeds, for 210 raw method–seed rows, and reports Wilson success intervals, paired per-seed success differences, bootstrap confidence intervals for mean selected-worst-error differences, and exact two-sided sign tests against all-windows continuous refinement and surrogate-QUBO simulated annealing. A separate focused QAOA-depth statistical package repeats the QUBO-derived methods over 30 deterministic seeds, for 150 raw method–seed rows, to isolate QAOA angle optimization and depth. The teacher-controlled experiment uses three deterministic seeds.

Unless otherwise stated, the thresholds are 0.09 for nominal success and 0.17 for selected robust recovery success. The robust-margin suite uses the same thresholds. It includes a thrust-margin selected-branch grid at phase time 0.2, $T = 0.5$ TU, and $N = 12$, and an all-one-segment-outage branch suite at $N = 8$, $a_{\max} = 0.3$, and $T = 0.5$ TU. In the latter suite, all-outage means all configured one-segment outage masks only; no two-segment outage branches are included. The fixed-final-time tail-coast package uses the same thresholds. Its combined `tail_coast_all_one_two_segment_t5_portfolio` row covers the configured one- and two-segment masks in one case row with `outage_lengths=[1,2]`, `outage_count=27`, and `selected_outage_count=27`; the separate all-single and all-two-segment rows remain provenance/scope rows. Its `meets_thresholds` flag is based on tail-coast nominal error and selected fixed-final-time worst error; optimizer convergence is reported separately so no-recovery-variable late-tail branches can be threshold-feasible without being counted as optimizer successes. The continuation-extension

Table 1: Catalog-derived pilot results. Values are medians across the generated pilot seeds. This benchmark did not meet the robust refinement thresholds.

Method	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
random	1.8309	2.5333	34.0442	0.3571	120.0000	0.0000	120.0000	0.0000
cross_entropy	0.9023	0.9031	9.9415	0.5000	120.0000	0.0000	120.0000	0.0000
genetic	2.5371	2.6132	34.8813	0.5000	132.0000	0.0000	132.0000	0.0000
true_sa	2.3540	2.4855	34.2025	0.6429	120.0000	0.0000	120.0000	0.0000
surrogate_qubo_sa	2.6497	2.7770	33.8472	0.5714	48.0000	72.0000	120.0000	0.0000
qaoa_statevector	2.4262	2.5573	33.8785	0.6429	48.0000	72.0000	120.0000	0.0000
all_windows_continuous	1.6383	2.2746	23.7862	1.0000	31.0000	0.0000	31.0000	0.0000

Table 2: Catalog-derived QUBO surrogate fit diagnostics.

seed	ridge	train_rmse	val_rmse	train_r2	val_r2	n_train	n_val	shared_qubo_training_evaluations	qubo_training_true_evaluations	equal_total_budget
11	0.0001	0.0002	4.3189	1.0000	0.5063	54	18	72	72	True
23	0.0001	0.0002	4.0060	1.0000	0.7190	54	18	72	72	True
37	0.0001	0.0002	5.9296	1.0000	0.0352	54	18	72	72	True

suite also uses these thresholds but is a continuous-backend direct multiple-shooting baseline: warm rows reuse persisted nominal controls from a source row, and the saved control sidecars record nominal controls, provenance, and hashes. The first Hermite–Simpson continuation baseline is a continuous-backend diagnostic with constant controls over each segment. The independent-midpoint-control Hermite–Simpson package adds separate midpoint control variables for nominal and selected recovery arcs; its sidecars persist both endpoint and midpoint nominal controls so warm-start provenance reproduces the reported trajectory and fuel diagnostics.

6. Results

6.1. Catalog-derived schedule benchmark remains challenging

Table 1 reports the catalog-derived pilot. QUBO simulated annealing and QAOA produce competitive pre-refinement schedules in this small run, but after fair accounting they require shared QUBO-training evaluations, and no method satisfies the configured post-refinement robust success thresholds. This is a negative result: a quantum-ready schedule sampler is not enough when the continuous recovery optimizer cannot turn the schedule into a feasible robust trajectory.

The QUBO diagnostics in Table 2 also show overfitting risk. With a full quadratic model and a small training set, the training fit is much stronger than the validation behavior. This motivates reporting QUBO fit quality as a first-order result rather than assuming that a learned quadratic model preserves the true trajectory ranking.

Table 3: Feasibility sweep stress test for the catalog-derived benchmark. None of the completed cases met both nominal and selected robust thresholds.

Transfer time	amax	Segments	Max nfev	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Meets thresholds	Best start	nfev	Runtime (s)
4.0000	0.2000	14	120	0.2008	0.2310	2.8975	False	feedback	251	1635.4279
4.0000	0.3000	14	150	0.1229	0.2783	7.5962	False	feedback	177	1102.4643
4.0000	0.3000	18	150	0.2423	0.2823	5.0379	False	feedback	39	351.0886
5.0000	0.3000	14	150	0.5810	0.6327	57.6216	False	zero	30	88.5080
5.0000	0.1500	14	120	0.6248	0.6785	53.6764	False	zero	138	997.2723
4.0000	0.2500	14	150	1.1558	1.2770	60.9139	False	feedback	81	324.7229
4.0000	0.1500	14	120	0.7389	1.6931	37.8087	False	zero	32	122.3204
3.0000	0.0650	14	120	1.0202	2.5174	7.1874	False	zero	61	326.9638

Table 4: Catalog-derived direct multiple-shooting diagnostics. Rows with zero selected outages are nominal-only diagnostics; all-outage diagnostic errors are still reported.

Transfer time	amax	Segments	Selected outages	Initialization	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Fuel	Max ||u||	Bound violation	nfev	Runtime (s)	Meets thresholds
4.0000	1.0000	14	1	blend	0.1714	0.0004	8.1180	4.0000	1.0000	0.0000	220	507.1263	False
5.0000	0.5000	14	0	linear	0.5154	0.5154	55.2845	2.5000	0.5000	0.0000	100	43.4353	False
4.0000	0.5000	14	3	linear	1.5355	1.2268	25.1426	2.0000	0.5000	0.0000	100	118.9605	False
4.0000	0.3000	14	3	linear	1.5981	1.9548	20.1304	1.2000	0.3000	0.0000	14	22.9601	False
4.0000	0.3000	14	3	linear	1.6273	4.3964	5.5503	1.2000	0.3000	0.0000	74	172.0316	False
5.0000	0.5000	14	3	linear	6.1753	5.7643	9.0650	2.5000	0.5000	0.0000	30	68.2684	False
6.0000	0.5000	14	0	linear	7.4830	7.4830	119.2412	3.0000	0.5000	0.0000	100	49.9072	False

6.2. Feasibility stress tests expose the refinement boundary

Table 3 shows the best completed points from the feasibility sweep. The best nominal error found in the sweep is 0.1229–0.2008 depending on the point, but selected outage errors remain above the 0.17 robust threshold and all-mask diagnostics remain much larger. Increasing thrust or transfer time does not monotonically fix the problem because the nonlinear dynamics and local least-squares convergence interact with the initialization.

The direct multiple-shooting diagnostics in Table 4 extend this stress test to seven completed catalog-derived rows. The high-thrust nominal-first selected-outage row, with $T = 4.0$ TU, $a_{\max} = 1.0$, $N = 14$, one selected outage, blend node initialization, and nominal-first homotopy enabled, drives the selected recovery worst error to 0.0004. It still fails the nominal threshold with nominal error 0.1714, retains an all-mask diagnostic worst error of 8.1180, and reaches the 220-evaluation cap with optimizer and backend success false. The best nominal-only row, $T = 5.0$ TU and $a_{\max} = 0.5$, reaches nominal error 0.5154 but has an all-mask diagnostic worst error of 55.2845. None of the rows satisfies the 0.09 nominal and 0.17 selected-recovery thresholds. The conservative interpretation is that selected branch success can coexist with nominal and all-mask diagnostic failure; this does not prove that multiple shooting is unsuitable, but it shows that the present sparse least-squares implementation and budget are insufficient for the hard catalog-derived case.

Table 5 reports the stronger targeted catalog attempt. Despite a larger thrust bound, multistart initialization, bang-bang feedback starts, and a 250-

Table 5: Targeted catalog-derived feasibility attempt with stronger multistart branch recovery. Thresholds were not relaxed.

Transfer time	amax	Segments	Max nfev	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Meets thresholds	Best start	nfev	Runtime (s)
4.0000	0.3000	14	250	0.3055	1.3513	5.4449	False	zero	92	389.4414

Table 6: Selected-outage hard-catalog feasibility-envelope diagnostics. Selected recovery branches are optimized for the listed masks, but all rows fail the nominal threshold and all-mask values are diagnostics, not robustness claims.

Case	Method	Transfer time	amax	Segments	Selected outages	Nominal error	Selected worst error	All-mask diagnostic worst error	Max $\|\|u\|\|$	Bound violation	nfev	Runtime (s)	Meets thresholds
selected_branch	multiple_shooting	4.0000	0.7000	14	1	0.3535	0.0360	21.8255	0.7000	0.0000	180	165.8788	False
selected_branch	multiple_shooting	4.0000	1.0000	14	1	0.8020	0.0009	9.3644	1.0000	0.0000	180	259.9800	False
selected_branch	multiple_shooting	4.0000	1.0000	14	3	1.6580	0.0150	8.7872	1.0000	0.0000	180	610.7961	False

evaluation limit, the best completed case misses both thresholds. The optimizer selected a low-control solution with nominal error 0.3055 and selected recovery worst error 1.3513. A $N = 18$ variant was terminated after more than nine minutes without a completed row. This negative result strengthens the interpretation that the present continuous backend is not a mature catalog-transfer optimizer.

Table 6 reports the selected-outage hard-catalog envelope. The selected recovery branches can be made small for the selected masks: the high-thrust one-outage row has selected-worst error 0.0009, the high-thrust three-outage row has selected-worst error 0.0150, and the mid-thrust one-outage row has selected-worst error 0.0360. These are not robust success cases. All three rows fail the nominal threshold, with nominal errors 0.8020, 1.6580, and 0.3535, respectively. All three hit the 180-evaluation cap, report optimizer/backend success as false, and retain high all-mask diagnostics of 9.3644, 8.7872, and 21.8255. The package is therefore useful negative evidence: selected recovery controls can overfit the chosen outage masks while the nominal trajectory and unselected outage family remain unresolved.

6.3. *Locked-nominal hard-catalog branch recovery narrows but does not close the blocker*

Table 7 and Figure 2 add a locked-nominal hard-catalog branch-recovery diagnostic on the same NRHO-like to DRO-phase benchmark at $a_{\max} = 1.0$, $N = 14$. Unlike the selected-outage envelope, this package first solves a single all-windows nominal multiple-shooting trajectory, then freezes that nominal control sequence. Each selected missed-thrust mask is optimized independently: controls before the outage are exactly the locked nominal controls with the missed segment(s) zeroed, and only the post-outage recovery controls are variables. The locked nominal trajectory itself reaches nominal

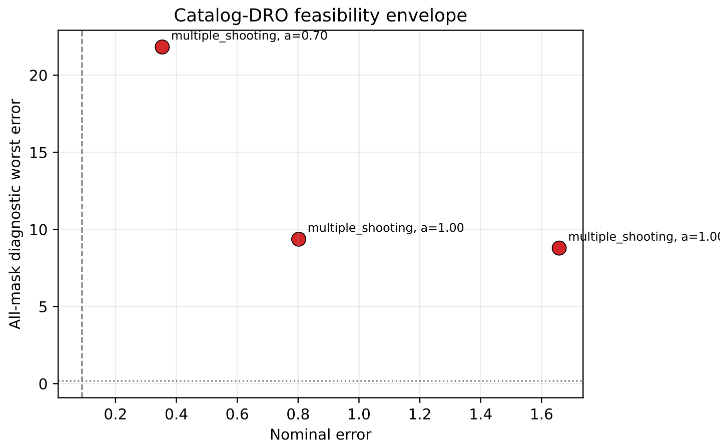

Figure 1: Selected-outage hard-catalog envelope. Small selected-outage errors do not imply robust success because nominal errors fail and all-mask diagnostics remain high.

error 0.0131, well below the 0.09 nominal threshold, so unlike the selected-outage envelope the nominal control is no longer the failing component. The all-mask column is a diagnostic: it evaluates every configured outage mask, using optimized recovery branches for selected masks and masked locked nominal controls for unselected masks, so it is not a robustness claim over the unselected family. This is a continuous-backend baseline, not QUBO, QAOA, or quantum evidence.

The row-level pattern is deliberately mixed. The `locked_hard_nominal_only` row runs no branch recovery, so its selected-worst error equals the nominal error 0.0131 by construction and it meets the thresholds only as a nominal-only diagnostic while the all-mask value stays at 8.5410. The `locked_hard_selected1` row optimizes the single hardest one/two-segment mask to selected-worst error 0.1973 and the `locked_hard_selected3` row optimizes the three hardest masks to the same worst error 0.1973; both exceed the 0.17 robust threshold and fail, and two of the three branches in the latter row hit the 120-evaluation cap with optimizer success false. The `locked_hard_all_single` row optimizes all 14 one-segment masks but the latest-outage masks leave too few recovery segments, so its selected-worst error is 0.8044 and it fails. The one positive row is `locked_hard_single_min6_selected8`: restricting the scope to one-segment outages with at least six remaining recovery segments and selecting

Table 7: Locked-nominal hard-catalog branch-recovery diagnostics on the NRHO-like to DRO-phase benchmark. The nominal control is solved once and frozen; each selected missed-thrust branch is optimized independently after the outage. The all-mask column is a diagnostic over every configured mask, not a robustness claim. The `locked_hard_single_min6_selected8` row meets the thresholds but is not optimizer-converged.

Case	Policy	Selected masks	Nominal error	Selected worst error	All-mask diagnostic worst	Max ‖u‖	Bound violation	nfev	Runtime (s)	Meets thresholds
locked_hard_all_single	all_single	14	0.0131	0.8044	8.5410	1.0000	0.0000	955	575.7630	False
locked_hard_nominal_only	0	0	0.0131	0.0131	8.5410	1.0000	0.0000	71	40.6265	True
locked_hard_selected1	1	1	0.0131	0.1973	8.1384	1.0000	0.0000	161	96.3449	False
locked_hard_selected3	3	3	0.0131	0.1973	6.2972	1.0000	0.0000	401	247.4372	False
locked_hard_single_min6_selected8	8	8	0.0131	0.1580	1.4982	1.0000	0.0000	823	495.8345	True

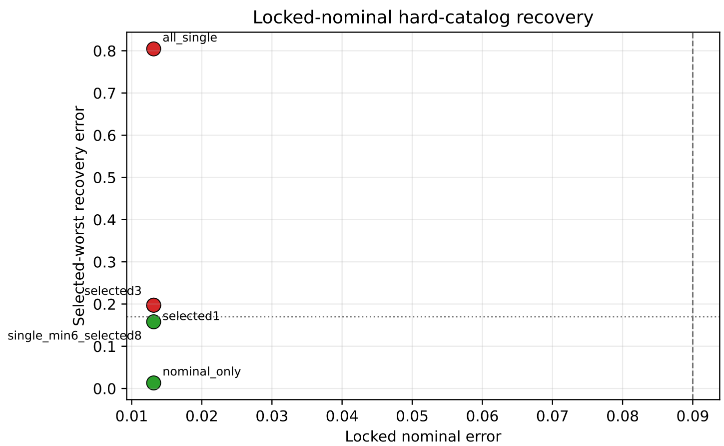

Figure 2: Locked-nominal hard-catalog branch recovery. With the nominal control held feasible, a bounded one-segment, six-recovery-segment selected subset meets the thresholds, while selected one/two-segment and all-single-outage scopes remain above the robust threshold.

the eight hardest eligible masks yields selected-worst error 0.1580 and all-mask diagnostic worst error 1.4982, meeting both thresholds. The caveat is explicit: `optimizer_success` is false for that row because several selected branches reached the 120-evaluation cap even though their terminal-error metrics already pass, so the row is reported as threshold-feasible but not optimizer-converged. The package therefore narrows the hard-catalog blocker—a bounded, early-enough one-segment outage scope is recoverable once the nominal control is feasible—without eliminating it, because the broader selected one/two-segment and all-single-outage scopes still fail.

6.4. Delayed-arrival hard-catalog portfolio recovery solves all one-segment masks

Table 8 and Figure 3 add a delayed-arrival variant of the locked-nominal hard-catalog diagnostic. The nominal all-windows control is still solved once to the original target at $T = 4.0$ TU and then frozen. For each selected outage branch, controls before `outage_end(mask)` are locked nominal controls with the missed segment zeroed. The branch optimizer then controls the remaining original segments plus h additional recovery segments of the same duration $\Delta t = T/N$. The delayed branch target is not the original target at T; it is the original target propagated ballistically by $h\Delta t$. Therefore these rows test delayed-arrival recovery horizon feasibility, not fixed-final-time robustness.

The three recorded rows deliberately separate provenance, a regularized negative, and the portfolio result. The nominal-only row uses $h = 4$ and runs no branch optimizer; it reports nominal original-target error 0.013133 and a delayed-coast selected metric 0.025525, while the all-mask delayed diagnostic remains 7.310180. The `delayed_hard_all_single_h4_regularized` row optimizes all 14 one-segment outage masks with the regularized branch residual and fails, with selected/all delayed worst error 1.398973. The `delayed_hard_all_single_h6_portfolio` row uses a six-segment arrival delay and a two-variant branch residual portfolio, `regularized_001` and `terminal_only`. Every variant is charged in the branch evaluation accounting, and the accepted branch candidate is selected by the recorded portfolio rule: optimizer-converged candidates that already meet the robust threshold are preferred, then ranked by terminal error and fuel. This h6 portfolio row optimizes all 14 one-segment hard-catalog outage branches and reaches selected/all delayed worst error 0.004040, nominal error 0.013133, maximum control norm 1.0, zero reported bound violation, `branch_optimizer_all_success=true`, 2253 total function evaluations, and runtime about 2639.7 s.

This is strong hard-catalog branch-recovery evidence, but the caveats are essential. The result is continuous-backend evidence only; it is not QUBO, QAOA, quantum, or schedule-sampling evidence. It is not fuel-optimal because the residual portfolio is a feasibility diagnostic and includes a terminal-only branch residual. It does not by itself solve fixed-final-time recovery, and it does not include two-segment outage families. Thus the nuanced hard-catalog conclusion from this package alone is that a six-segment delayed-arrival horizon with an explicitly charged branch residual portfolio recovers

Table 8: Delayed-arrival locked-nominal hard-catalog branch-recovery diagnostics. Branch duration is $(N+h)\Delta t$ and the branch target is the original target propagated ballistically by $h\Delta t$. The h6 portfolio row optimizes all 14 one-segment masks, charges all residual variants, and is delayed-arrival feasibility evidence only.

Case	Policy	Selected masks	Horizon segments	Portfolio	Variants	Nominal original error	Nominal delayed coast error	Selected delayed worst error	All-mask delayed diagnostic worst	Terminal weight	Control weight	Smooth weight	Continuity weight	Max ||u||	Bound violation	nfev	Runtime (s)	Meets thresholds	Nominal optimizer success	Branch optimizer ran	Branch optimizer all success
delayed_hard_all_single_h4_regularized	all_single	14	4	False	["configured"]	0.0131	0.0255	1.3990	1.3990	4.0000	0.0100	0.0100	0.0000	1.0000	0.0000	1338	1076.4599	False	True	True	False
delayed_hard_all_single_h6_portfolio	all_single	14	6	True	["regularized_001", "terminal_only"]	0.0131	0.0380	0.0040	0.0040	4.0000	0.0100	0.0100	0.0000	1.0000	0.0000	2253	2639.7199	True	True	True	True
delayed_hard_nominal_only_h4	0	0	4	False	["configured"]	0.0131	0.0255	0.0255	7.3102	4.0000	0.0100	0.0100	0.0000	1.0000	0.0000	71	37.4396	True	True	False	False

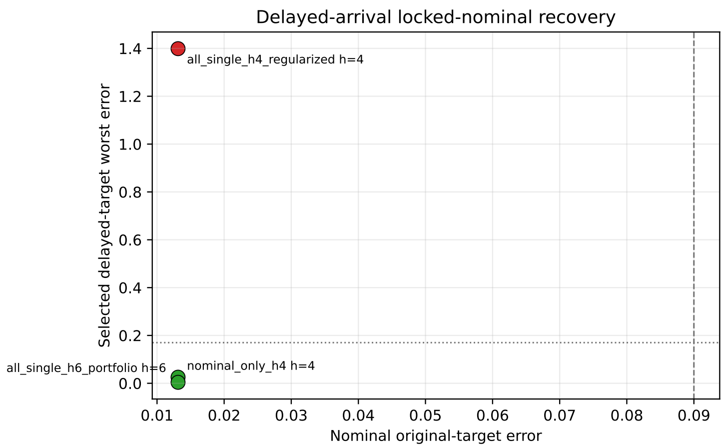

Figure 3: Delayed-arrival locked-nominal hard-catalog recovery. The h4 regularized all-single row fails, while the h6 branch-residual portfolio recovers all one-segment masks against the delayed target.

every one-segment outage mask in the recorded hard catalog-DRO case.

6.5. Tail-coast fixed-final-time thresholds over configured masks

Table 9 reports the fixed-final-time tail-coast hard-catalog diagnostic. The setup again solves and freezes a locked nominal control, but then refines it with the final five nominal control segments constrained to exact zero. This mixed-regularized tail nominal uses control and smoothness weights of 0.005 and a 220-evaluation tail-nominal budget. Branches keep the original target and original final time; no delayed target or added recovery horizon is used. The nominal-only provenance row has tail-coast nominal error 0.022992 and therefore meets the threshold rule as nominal provenance, but with no selected branch recovery its masked-nominal all-mask diagnostic remains 27.8351.

The main result is now the combined `tail_coast_all_one_two_segment_t5_portfolio` row. It sets `outage_lengths=[1,2]`, selects all configured masks

Table 9: Fixed-final-time tail-coast locked-nominal hard-catalog recovery. The combined row selects/evaluates all 27 configured one- and two-segment masks in one fixed-final-time case row; separate all-single and all-two-segment rows are retained as provenance/scope rows. Branches with post-outage recovery variables are optimized. No-recovery-variable late-tail branches are direct threshold evaluations, so false branch optimizer all-success flags are accounting caveats rather than threshold failures.

Case	Policy	Outage lengths	Selected masks	Tail coast segments	Portfolio variants	Fallback eval branches	Accepted fallback branches	No-recovery branches	Tail-coast nominal error	Selected fixed-time worst error	All-mask fixed-time diagnostic worst	Max $\|\|u\|\|$	Bound violation	nfev	Meets thresholds	Branch optimizer ran	Accepted branch optimizers converged
tail_coast_nominal_only_t5	0	[1]	0	5	1	0	0	0	0.0230	0.0230	27.8351	1.0000	0.0000	287	True	False	False
tail_coast_all_single_t5_portfolio	all_single	[1]	14	5	2	1	1	1	0.0230	0.0230	0.0230	1.0000	0.0000	2057	True	True	False
tail_coast_all_two_segment_t5_portfolio	all_configured	[2]	13	5	2	3	3	1	0.0230	0.0936	0.0936	1.0000	0.0000	3467	True	True	False
tail_coast_all_one_two_segment_t5_portfolio	all_configured	[1, 2]	27	5	2	4	4	2	0.0230	0.0936	0.0936	1.0000	0.0000	5929	True	True	False

with `selected_outage_policy=all_configured`, and evaluates all 27 selected masks in one fixed-final-time case row. The row reaches tail-coast nominal error 0.02299233817855882, selected fixed-time worst error 0.0936063931709301, and all-mask fixed-time diagnostic worst error 0.0936063931709301, so it meets the 0.09/0.17 threshold rule. Its branch portfolio evaluates and accepts deterministic fallback starts for four branches, charges 5929 function evaluations, and records runtime about 1833.6 s in the current CSV. The branch-recovery accounting contains the one-segment recovery-count sequence $[13, \ldots, 0]$ and the two-segment sequence $[12, \ldots, 0]$, so two late-tail branches have no post-outage recovery variables. Those two branches are direct fixed-final-time threshold evaluations, not optimizer-converged branch solves; consequently `branch_optimizer_all_success=false` even though `meets_thresholds=true`.

The separate `tail_coast_all_single_t5_portfolio` and `tail_coast_all_two_segment_t5_` rows remain useful provenance rows. The all-single row selects all 14 one-segment masks and reaches selected/all fixed-final-time worst error 0.02299233817855882; it evaluates fallbacks only for mask 7, accepting `constant_y_minus_0p5`, and has one no-recovery-variable late-tail mask. The all-two-segment row selects all 13 configured two-segment masks and reaches selected/all fixed-final-time worst error 0.0936063931709301; it evaluates accepted fallbacks `constant_y_minus_0p5`, `constant_x_minus_0p5`, and `constant_x_plus_0p5` for masks 0, 6, and 7, and has one no-recovery-variable late-tail mask. The combined row supersedes the earlier split-scope interpretation for the configured one/two fixed-final-time tail-coast diagnostic. It does not establish fuel optimality, a production-grade optimal-control solution, high-fidelity validation, quantum advantage, or robustness to outage families beyond the configured one- and two-segment masks in this locked-nominal continuous-backend setup.

Because these results mix negative sampler evidence, continuous-backend feasibility diagnostics, and scoped hard-catalog recovery rows, Table 10 records the claim hierarchy used for the rest of the manuscript.

6.6. Bounded projected multiple shooting solves limited non-teacher phase-shift cases

Table 11 reports the compact bounded-projected multiple-shooting benchmark on the catalog halo phase-shift target. The positive row uses $N = 12$, $a_{\max} = 0.2$, one selected outage branch, linear node initialization, and a 40-evaluation cap. It reaches nominal error 0.0479, selected recovery worst error 0.0619, and all-mask diagnostic worst error 0.0752, all below the configured thresholds. The reported maximum acceleration norm is 0.2, and the largest bound violation is 5.6×10^{-17}, i.e., numerical roundoff.

This result is a backend result rather than a quantum-sampling result. The selected outage branch, index 6 in the saved metadata, is optimized with branch recovery. The all-mask diagnostic evaluates every one-segment outage; unselected outages use the nominal controls with the outage mask applied, not separately optimized recovery branches. The diagnostic is therefore stricter than selected-branch success in one sense, because every unselected one-segment outage remains below threshold without its own recovery solve, but it is not the same as optimizing all outage branches. The rows selecting all one-segment outage branches at $N = 8$ and $N = 12$ fail the robust threshold under their smaller completed budgets, with selected/all-mask worst errors 0.3378 and 0.3904, respectively. Thus the bounded projected backend fixes a real control-bound inconsistency and provides a limited non-teacher positive result, but it does not close the hard catalog-transfer gap.

Table 12 extends this backend diagnostic across phase times 0.2, 0.3, and 0.4 with row-level settings fingerprints, configuration hashes, and source-state identifiers so that resumed rows cannot be silently reused after a settings change. Two of the three selected-branch rows meet the thresholds. The phase-time 0.4 row is the strongest completed case, with nominal error 0.0260, selected recovery worst error 0.0083, and all-mask diagnostic worst error 0.0274. The phase-time 0.2 row fails both thresholds, and an all-outage-selected diagnostic at phase time 0.3 fails with selected/all-mask worst error 0.3581. The broader suite therefore strengthens the continuous-backend evidence while preserving the main limitation: selected-branch feasibility over one outage does not imply successful simultaneous optimization over every outage branch.

Table 13 adds a separate bounded projected trapezoidal direct-collocation baseline on the same phase-shift suite. The direct-collocation variables include nominal state nodes, nominal controls, selected outage branch nodes, and selected branch recovery controls; the residual enforces trapezoidal dynamics defects and terminal conditions while projecting every reported and evaluated control vector to $\|u_i\| \leq a_{\max}$. This baseline succeeds only for phase time 0.4, with nominal error 0.0357, selected recovery worst error 0.0191, all-mask diagnostic worst error 0.0602, and maximum control norm 0.1483. It fails at phase times 0.2 and 0.3 under the compact 30-evaluation budget. The result addresses the missing-continuous-baseline concern in a limited way: a recognizable direct transcription agrees that the easier phase-time 0.4 case is solvable, but the compact implementation is not a mature production collocation or continuation workflow and does not solve the harder phase-shift rows.

Table 14 adds the robust-margin suite. In the selected-branch thrust-margin grid at phase time 0.2, $T = 0.5$ TU, and $N = 12$, only one of the 12 one-segment outage masks is optimized. At $a_{\max} = 0.20$, the row has nominal error 0.193922, selected worst error 0.216827, and all-mask diagnostic worst error 0.228764, so it fails both thresholds. At $a_{\max} = 0.25$, the row has nominal error 0.113567, selected worst error 0.141845, and all-mask diagnostic worst error 0.154253; it fails only because the nominal error remains above 0.09. At $a_{\max} = 0.30$, the row passes with nominal error 0.053774, selected worst error 0.073781, and all-mask diagnostic worst error 0.086309. This isolates phase-time 0.2 as thrust-margin sensitive rather than categorically infeasible for the bounded-projected backend.

The same table also optimizes all configured one-segment outage branches in a smaller $N = 8$, $a_{\max} = 0.3$ suite. Here, all-outage refers only to all one-segment outage masks; two-segment outage families are not part of this suite. The phase-time 0.2 row has nominal error 0.047231 but selected/all-mask worst error 0.252269, so it fails the robust threshold. The phase-time 0.3 row has nominal error 0.055472 and selected/all-mask worst error 0.035816, so it is threshold-feasible, but it reached the 60-evaluation limit and is not reported as optimizer-converged. The phase-time 0.4 row passes with nominal error 0.053086 and selected/all-mask worst error 0.014963. These rows partially address the selected-branch-only critique by showing all one-segment outage branch optimization in smaller higher-thrust cases, while leaving the hard catalog-DRO benchmark and larger outage families unresolved.

Table 15 adds the continuation-extension continuous-backend package.

Warm rows reuse persisted nominal controls from a source row as direct multiple-shooting warm starts, and the control sidecars store the nominal controls together with provenance and hashes. This is not a quantum, QUBO, QAOA, or discrete schedule-sampling result.

For the $N = 8$ all-single-outage rows, all eight one-segment outage masks are selected for optimization. The phase-time 0.3 cold row has nominal error 0.055474 and selected/all-mask worst error 0.035126, and the phase-time 0.4 row warm-started from the phase-time 0.3 nominal controls has nominal error 0.053098 and selected/all-mask worst error 0.013913. Both rows satisfy the thresholds with optimizer and backend success. For one- and two-segment outage masks, the extension suite selects all configured masks. The $N = 6$ phase-time 0.3 cold row selects all 11 masks and meets thresholds with nominal error 0.059163 and selected/all-mask worst error 0.069048. The $N = 8$ phase-time 0.3 cold row selects all 15 one- and two-segment masks and meets thresholds with nominal error 0.061201 and selected/all-mask worst error 0.043491. The remaining $N = 6$ phase-time 0.2 row, warm-started from the phase-time 0.3 nominal controls and allowed 160 function evaluations, has nominal error 0.065112 but selected/all-mask worst error 0.219306, so it fails the robust threshold and reports optimizer/backend failure.

The interpretation is deliberately limited. The extension suite gives four optimizer-converged feasible continuous-backend rows out of five and broadens all-selected outage evidence to one- and two-segment masks at $N = 8$, phase time 0.3. It does not solve the phase-time 0.2 one/two-segment diagnostic, does not solve the hard NRHO-like to DRO-phase benchmark, does not prove quantum advantage, and reinforces that continuous-control initialization can dominate schedule sampling.

6.7. Constant-control Hermite–Simpson continuation probe narrows but does not close the backend gap

Table 16 and Figure 4 add a constant-control Hermite–Simpson direct-collocation continuation baseline/probe on the catalog halo phase-shift target family. Warm rows reuse persisted nominal controls from the phase-time 0.3 cold row, giving trajectory-stacking semantics similar in spirit to collocation continuation workflows. The qualifier constant-control is important: the transcription uses Hermite–Simpson state defects, but controls are held constant over each segment and no independent midpoint control variables are optimized. This is a continuous-backend diagnostic, not QUBO, QAOA, discrete, or quantum evidence.

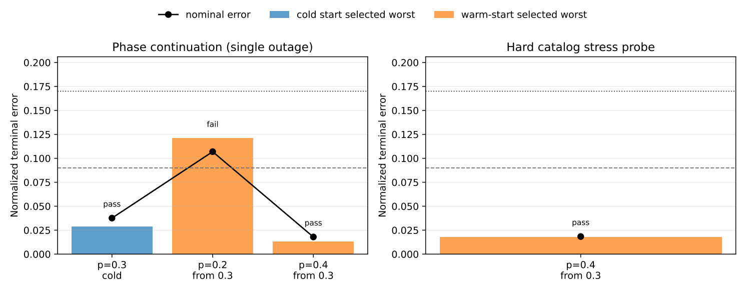

Figure 4: Constant-control Hermite–Simpson continuation baseline/probe summary. Threshold-feasible phase-shift rows remain continuous-backend diagnostics and should not be read as quantum or hard-catalog transfer success.

The row-level pattern is useful because it is not uniformly positive. The phase-time 0.3 cold row meets the nominal and selected-worst thresholds with nominal error 0.0376 and selected worst error 0.0287, but it stops at the 50-evaluation cap and therefore reports optimizer success false. The phase-time 0.2 warm row fails the nominal threshold, with nominal error 0.1069, even though the selected-worst error is 0.1208. The phase-time 0.4 warm row meets both thresholds with optimizer success true after seven function evaluations. The `hs_hard_p04_amax02_warm_from_p03` row also meets both thresholds at $a_{\max} = 0.2$ with optimizer success true, but its target mode is still `catalog_halo_phase_shift`. It is a lower-thrust phase-shift stress probe, not the unresolved hard NRHO-like to DRO-phase catalog benchmark.

The interpretation is therefore conservative. Constant-control Hermite–Simpson continuation provides another independently parameterized continuous-backend check and confirms that the easier phase-time 0.4 phase-shift target is not an artifact of one multiple-shooting implementation. It does not solve the phase-time 0.2 diagnostic, does not optimize all one- and two-segment outage families, and does not change the negative conclusion for the hard catalog-derived benchmark.

6.8. Independent-midpoint-control Hermite–Simpson closes the midpoint-control transcription critique

Table 17 and Figure 5 add an independent-midpoint-control Hermite–Simpson direct-collocation evidence package. Each segment optimizes an endpoint/segment control and a separate midpoint control; the Hermite–Simpson defect uses the endpoint control in f_{left} and f_{right} and the independent midpoint control in f_{mid}. Reporting propagation uses RK4 substeps with quadratic endpoint–midpoint–endpoint control interpolation. The nominal-control sidecars persist both endpoint and midpoint controls, with separate endpoint, midpoint, and combined sidecar hashes, so warm-start rows can reproduce the reported midpoint-control trajectory, fuel, and terminal-error diagnostics.

The result directly addresses the specific criticism that the earlier Hermite–Simpson probe was constant-control and had no independent midpoint controls, but it does not turn the continuous evidence into quantum or discrete-sampler evidence. Three rows meet the configured thresholds: the phase-time 0.3 cold row, the phase-time 0.4 warm row, and the lower-thrust phase-time 0.4 stress row. The phase-time 0.2 warm row remains unresolved, with nominal error 0.0976 above the 0.09 nominal threshold even though selected-worst error is below the robust threshold. The true catalog-DRO selected-outage diagnostic is retained as a negative row: nominal error 4.2644, selected-worst error 11.0843, optimizer/backend success false, and 50 function evaluations. Thus independent midpoint controls strengthen the continuous baseline and close the midpoint-control objection, while preserving the main scientific caveats.

6.9. Non-teacher phase-shift benchmark is solved only by the continuous all-windows baseline

Table 18 reports the catalog halo phase-shift benchmark. This case removes the teacher-generation objection: the target is a zero-thrust CR3BP phase point of the JPL L2 southern halo source state, and the saved metadata records `teacher_generated=false`. The all-windows continuous baseline succeeds in all five seeds, with median nominal error 0.0418, median selected recovery worst error 0.0750, and median all-mask diagnostic worst error 0.0887. The result also exposes a different limitation: none of the tested binary schedule samplers finds a schedule that refines to the thresholds in this five-seed run. QAOA gives the best median among the discrete initializers, but its median selected recovery error remains 0.2625, above the 0.17 threshold.

The phase-shift QUBO surrogate is not the limiting factor in the same way as in the teacher benchmark. Across five seeds, validation R^2 ranges from 0.321 to 0.955 and validation Spearman correlation ranges from 0.310 to 1.000. The remaining failure is therefore more structural: the robust binary objective and the tested samplers prefer sparse late-thrust schedules such

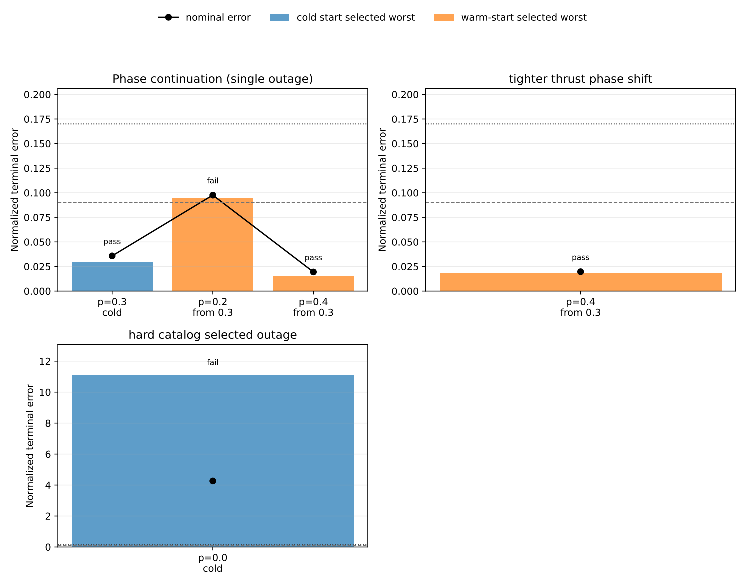

Figure 5: Independent-midpoint-control Hermite–Simpson continuation summary. The package strengthens the continuous-backend comparison but keeps the phase-time 0.2 and catalog-DRO diagnostics unresolved.

as `00000111` or `00001111`, while the successful baseline uses all thrust windows and lets the continuous optimizer shape the acceleration. In this case, the evidence favors continuous-control freedom over sparse binary schedule selection.

6.10. Dense schedule repair exposes the sparsity bottleneck

Table 19 reports a deterministic schedule-repair ablation on the same non-teacher phase-shift benchmark. The repair is `prefix_to_last_active`: before continuous refinement, every window through the last active solver window is made available for thrust, while later zeros are preserved. This is a heuristic dense-availability envelope, not an oracle continuous control and not a quantum routine.

With repair enabled, every tested initializer succeeds in 5/5 seeds and attains the same median refined errors as the all-windows continuous baseline: nominal error 0.0418, selected recovery worst error 0.0750, and all-mask diagnostic worst error 0.0887. The provenance columns are the important result. Repaired methods enter refinement with sparse candidate active fractions, typically 0.375 and 0.500, but every method refines the same dense schedule `11111111` with refined active fraction 1.0. Therefore, this ablation does not rank the samplers and does not establish a quantum advantage. It shows that the current binary objective can place thrust late enough for a deterministic dense repair to recover a useful all-windows envelope, while the unrepaired binary encoding is too sparse for the continuous backend.

6.11. Duty-cycle prior yields a positive non-teacher discrete-initialization result

Table 20 reports the $N = 12$ phase-shift benchmark with the active-fraction prior J_ρ set to $\rho = 11/12$. This is the first non-teacher case in the paper where the discrete layer contributes a useful refinement structure rather than only diagnosing failure. Cross-entropy, genetic search, true-objective simulated annealing, and surrogate-QUBO simulated annealing each succeed in 10/10 seeds and select one-coast schedules with median refined active fraction 0.9167. Random search succeeds in 2/10 seeds and QAOA succeeds in 4/10 seeds under the same total true-evaluation budget.

The result is a tradeoff, not a dominance claim. The all-windows continuous baseline also succeeds in 10/10 seeds and has lower terminal errors: median nominal error 0.0386 and selected recovery worst error 0.0601. The one-coast methods have higher but feasible errors, with surrogate-QUBO simulated annealing at median nominal error 0.0690 and selected recovery worst error 0.0941. Their apparent benefit against the default all-windows setting is lower nominal fuel: 0.0917 versus 0.1000, an 8.33% reduction in this normalized fixed-time metric. The QUBO surrogate is well aligned with the modified objective in this benchmark, with mean validation $R^2 = 0.969$, mean validation Spearman correlation 0.962, mean pairwise order accuracy 0.935, and mean top-k recall 0.86 across ten seeds. Thus, the evidence supports a narrow claim: a duty-cycle-aware QUBO objective can recover a refinable high-duty schedule class on a non-teacher cislunar benchmark, but strong classical heuristics match it and the initially tested shallow random-angle QAOA sampler does not.

6.12. Thirty-seed main-method statistics favor all-windows error over sampler ranking

Tables 21 and 22, together with Figure 6, repeat the configured $N = 12$ duty-cycle-prior method comparison over 30 deterministic seeds. The raw package contains 210 rows, i.e., 30 seeds times seven methods: random search, cross-entropy search, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, the configured statevector QAOA sampler, and all-windows continuous refinement.

The success-rate evidence strengthens the 10-seed result but does not create a quantum-performance claim. Random search succeeds in 11/30 seeds, with Wilson 95% interval $[0.219, 0.545]$. Cross-entropy, genetic search, true-objective simulated annealing, surrogate-QUBO simulated annealing, and all-windows continuous refinement each succeed in 30/30 seeds, with Wilson interval $[0.886, 1.000]$. The configured `qaoa_statevector` method succeeds in 13/30 seeds, with interval $[0.274, 0.608]$.

The paired error comparisons are more important for the aerospace interpretation. Against all-windows continuous refinement, every sampled method has a positive mean selected-worst-error difference; negative would favor the sampled method. The exact two-sided sign-test value is approximately 1.86×10^{-9} for each sampler versus all-windows continuous refinement, even though the table rounds this to 0.0000. Thus, the all-windows continuous baseline is not just equally successful for the strongest methods; it also gives lower selected-worst error on this benchmark. Against surrogate-QUBO simulated annealing, cross-entropy, genetic search, and true-objective simulated annealing tie the success rate and have small nonsignificant selected-worst-error deltas. Random search is worse. The configured statevector QAOA row has lower success and a positive mean selected-worst-error delta of 0.0153 versus surrogate-QUBO simulated annealing, with exact sign-test $p = 0.1849$.

This package therefore strengthens the statistics beyond a QAOA-only ablation while preserving the conservative conclusion. Duty-cycle priors help several samplers find feasible high-duty schedules, but all-windows continuous fuel regularization remains a stronger mechanism for selected-worst error in this benchmark, and the evidence does not support quantum advantage or QAOA superiority.

6.13. Thirty-seed QAOA-depth statistics improve QAOA but do not support superiority

Tables 23 and 24 isolate the QUBO-sampling step on the same $N = 12$ duty-cycle-prior benchmark with 30 deterministic seeds. Each QUBO-derived method uses the same accounting per seed: 96 shared trajectory evaluations to train the QUBO and 24 true post-QUBO candidate evaluations. The resulting raw package contains 150 method–seed rows.

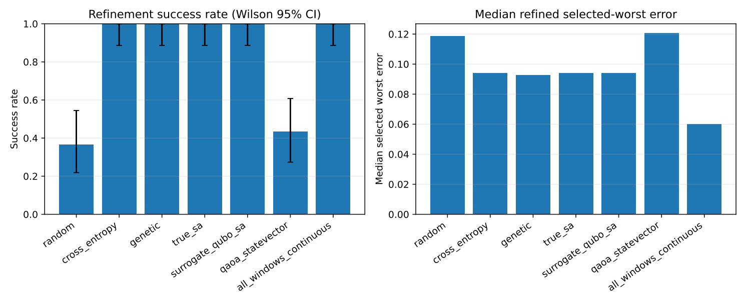

Figure 6: Thirty-seed main-method statistics for the duty-cycle-prior benchmark. High-duty samplers can be feasible, but all-windows continuous refinement remains lower-error on paired selected-worst comparisons.

The 30-seed success rates with Wilson 95% intervals are $28/30 = 0.933$ $[0.787, 0.982]$ for surrogate-QUBO simulated annealing, $28/30 = 0.933$ $[0.787, 0.982]$ for exact evaluation of the top 24 QUBO bitstrings, $15/30 = 0.500$ $[0.332, 0.668]$ for random-angle $p = 1$ QAOA, $21/30 = 0.700$ $[0.521, 0.833]$ for optimized $p = 1$ QAOA, and $29/30 = 0.967$ $[0.833, 0.994]$ for optimized $p = 2$ QAOA. Optimized $p = 2$ is competitive with the strongest classical QUBO sampler in this narrow statevector QUBO ablation, but it is not statistically supported as superior: versus surrogate-QUBO simulated annealing, its paired success delta is only $+0.033$, the mean selected-worst-error delta is -0.00056 with bootstrap 95% confidence interval $[-0.00393, 0.00236]$, and the exact sign-test p-value is 0.4545.

The useful positive result is narrower. Relative to random-angle $p = 1$ QAOA, optimized $p = 2$ QAOA increases success by $+0.467$ and reduces the mean selected-worst error by 0.0125 with bootstrap interval $[-0.0188, -0.0060]$. The exact sign-test p-value is still not significant at conventional levels ($p = 0.2478$), so the evidence should be read as practical improvement in this ablation rather than a formal dominance claim. No quantum advantage is claimed.

6.14. Cardinality ablation and fuel-error Pareto check

Table 25 places the duty-cycle result in context by exhaustively refining high-duty schedule classes. All 12 one-coast schedules are feasible, and 41 of 66 two-coast schedules are feasible. This means the duty-cycle prior is not discovering a rare one-coast structure; it is steering the search into a broad feasible high-duty region. Within the one-coast class, surrogate-QUBO

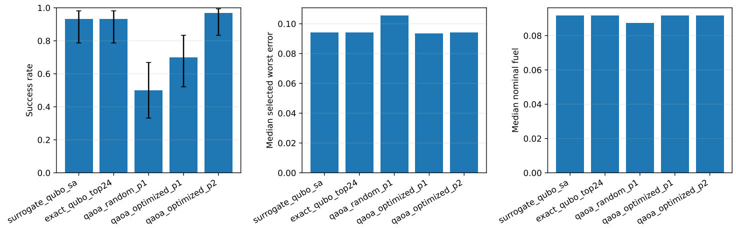

Figure 7: Thirty-seed QAOA-depth ablation summary. Optimized $p = 2$ QAOA improves over random-angle QAOA and is competitive with classical QUBO sampling, but paired tests do not support a superiority claim over surrogate-QUBO simulated annealing.

simulated annealing selects feasible schedules in 10/10 runs, but none of those schedules lies on the one-coast Pareto frontier used by the ablation, and its median selected-worst gap to the best one-coast schedule is 0.0108.

Table 26 then checks whether the one-coast fuel reduction survives a stronger continuous all-windows baseline. It does not. Increasing the all-windows fuel residual weight to 0.05 keeps the trajectory feasible, with nominal error 0.0773, selected recovery worst error 0.1100, all-mask worst error 0.1122, and nominal fuel 0.0846, lower than the one-coast median fuel 0.0917. The Pareto interpretation is therefore that binary duty-cycle control finds a feasible high-duty schedule class, while continuous fuel regularization inside all available windows is currently more effective at trading terminal error for fuel in this benchmark.

6.15. Teacher-controlled benchmark demonstrates selected-branch recovery

Table 27 reports the teacher-controlled benchmark. Random search succeeds in 2/3 seeds. Cross-entropy, surrogate QUBO simulated annealing, and QAOA each succeed in 1/3 seeds. Genetic search, true-objective simulated annealing, the all-windows continuous baseline, and the teacher-seeded binary schedule do not meet the selected-branch thresholds in this three-seed run. The teacher-controls oracle diagnostic succeeds in 3/3 seeds, with median nominal error 0.0067 and selected recovery worst error 0.0080, but it is not a valid competing initializer because it receives the hidden continuous controls used to generate the benchmark target.

The result has two interpretations. The positive interpretation is that the proposed architecture can produce selected-branch robust refinements on a feasible low-thrust transfer, including through QUBO and QAOA initializers. The limiting interpretation is more important for publication: random search

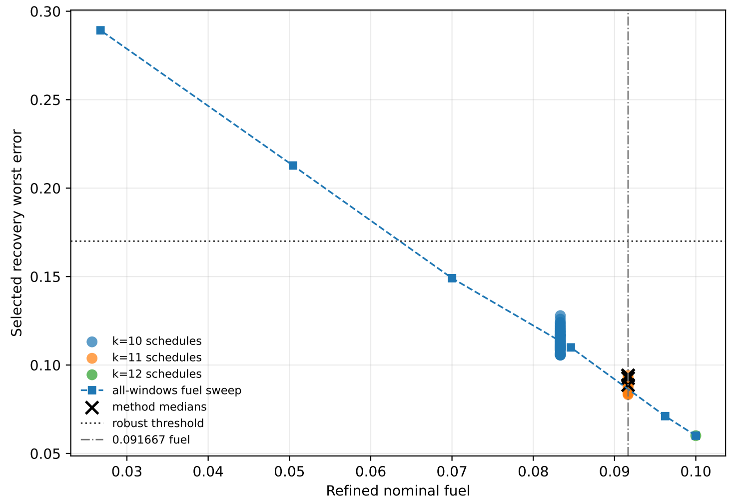

Figure 8: Fuel-error Pareto view for the duty-cycle-prior ablation. Lower fuel is achievable with all-windows fuel regularization while retaining feasibility, so the one-coast result should be read as a schedule-structure feasibility result rather than a fuel-optimality result.

currently has the highest non-oracle success rate, all-mask diagnostic errors remain much larger than selected-branch errors, and the oracle result shows that continuous-control initialization can dominate the outcome. Therefore the present data support feasibility of the pipeline and diagnose a bottleneck, not superiority of quantum-ready sampling.

The seed counts outside the 30-seed packages are still small. Wilson 95% intervals are wide: the phase-shift all-windows success rate of $5/5$ has interval approximately $[0.57, 1.00]$, the unrepaired phase-shift discrete-method rates of $0/5$ have intervals approximately $[0, 0.43]$, and the repaired phase-shift method rates of $5/5$ again have intervals approximately $[0.57, 1.00]$ but are method-collapsed by the deterministic repair. In the legacy 10-seed cardinality-prior comparison, $10/10$ rates have intervals approximately $[0.72, 1.00]$, random search at $2/10$ has interval approximately $[0.06, 0.51]$, and the shallow QAOA row at $4/10$ has interval approximately $[0.17, 0.69]$. The 30-seed main-method package narrows these intervals for the duty-cycle-prior comparison: random search is $11/30$ with interval $[0.219, 0.545]$, the configured statevector QAOA row is $13/30$ with interval $[0.274, 0.608]$, and the strongest classical high-duty samplers plus all-windows continuous refinement are $30/30$ with interval $[0.886, 1.000]$. The 30-seed QAOA-depth

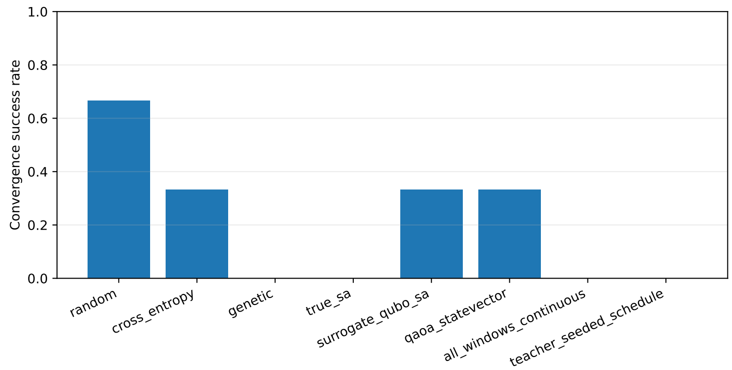

Figure 9: Teacher-controlled selected-branch refinement success rates. The result demonstrates nonzero recovery success but not quantum advantage.

package improves statistical rigor for the QUBO-derived samplers, but even there paired comparisons do not show statistically supported superiority of optimized $p = 2$ QAOA over surrogate-QUBO simulated annealing. The teacher random-search rate of 2/3 has interval approximately $[0.21, 0.94]$, and the teacher QUBO/QAOA rates of 1/3 have intervals approximately $[0.06, 0.79]$. These intervals are reported to prevent overinterpretation; they do not support a statistically significant ranking of all stochastic initializers or any quantum-advantage claim.

Table 28 reports nominal and recovery fuel. QAOA has the lowest median fuel among methods with nonzero recovery success, but its selected robust error and success rate are worse than random search. Fuel therefore cannot be evaluated independently of feasibility.

6.16. Surrogate quality remains a bottleneck

Table 29 shows teacher-controlled QUBO diagnostics. Validation R^2 varies from negative to moderately positive across only three seeds. Rank diagnostics are more favorable but still not decisive: validation Spearman correlation is 0.57–0.71, pairwise order accuracy is 0.68–0.79, and top-2 recall is 0.5–1.0 on only eight validation samples. This is too weak for a final surrogate-generalization claim and likely contributes to the lack of QUBO/QAOA dominance. A submission-ready version needs larger training sets, cross-validated regularization, ranking-oriented validation, or a structured surrogate with trajectory-informed features.

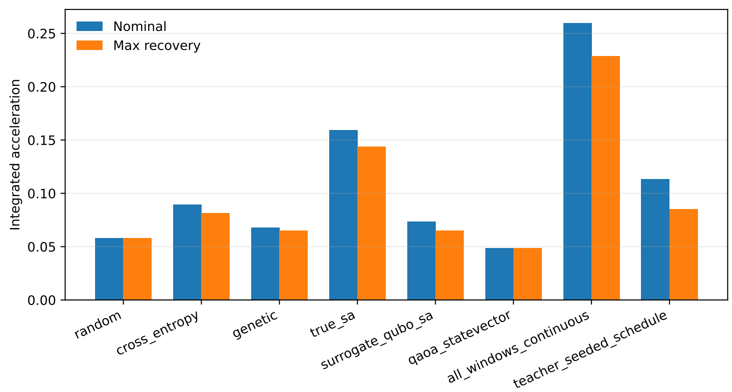

Figure 10: Teacher-controlled nominal and recovery fuel summaries. Fuel differences must be interpreted together with feasibility and missed-thrust error.

7. Discussion

The experiments support fifteen conclusions. First, robust low-thrust initialization can be represented as a binary scheduling problem with trajectory-evaluated missed-thrust penalties and then approximated by a QUBO. Second, the phase-shift benchmark shows that the present branch-recovery backend can solve limited non-teacher catalog/dynamics-derived cases under the configured thresholds, although most positive rows still optimize selected outage branches and report unselected masks diagnostically. Third, enforcing the acceleration bound inside the multiple-shooting residual removes an important implementation inconsistency and produces limited positive non-teacher backend results. Fourth, the robust-margin suite partially addresses the selected-branch-only criticism: in the smaller higher-thrust $N = 8$, $a_{\max} = 0.3$ setting, all one-segment outage branches are optimized and the phase-time 0.3 and 0.4 rows meet the thresholds, while the non-continuation phase-time 0.2 all-single-outage row remains robust-infeasible. Fifth, continuation extension diagnostics give four optimizer-converged feasible continuous-backend rows out of five, including all-selected $N = 8$ one- and two-segment masks at phase time 0.3, but the $N = 6$ one/two-segment phase-time 0.2 diagnostic still fails the robust threshold. Sixth, the constant-control Hermite–Simpson continuation probe provides another continuous-backend check: phase-time 0.4 warm continuation and a lower-thrust phase-shift stress row converge and meet thresholds, while the phase-time 0.2 warm row fails the nominal threshold

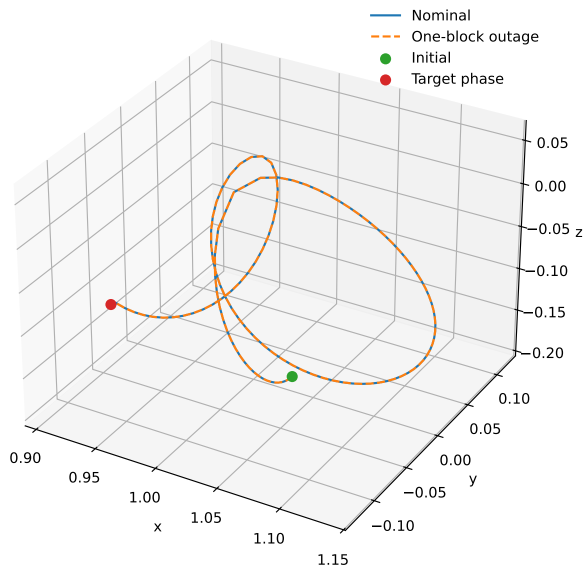

Figure 11: Example trajectory from the teacher-controlled benchmark. The case is a normalized CR3BP algorithm benchmark, not a flight-ready trajectory.

and the phase-time 0.3 cold row reaches the evaluation cap despite meeting thresholds. Seventh, the independent-midpoint-control Hermite–Simpson package closes the specific no-independent-midpoint-control critique: it optimizes midpoint controls, persists endpoint and midpoint sidecars, and keeps three threshold-feasible phase-shift rows, but it still fails the phase-time 0.2 nominal threshold and the bounded catalog-DRO selected-outage diagnostic. Eighth, the phase-time 0.2 thrust-margin grid shows clear sensitivity to available thrust: the selected-branch row fails at $a_{\max} = 0.20$, fails by nominal error only at $a_{\max} = 0.25$, and passes at $a_{\max} = 0.30$. Ninth, a separate bounded trapezoidal direct-collocation baseline succeeds on the easiest phase-suite row and fails on the two harder rows, supporting the interpretation that backend capability is strongly case-dependent. Tenth, that same phase-shift benchmark shows that the tested binary schedule samplers can fail even when the continuous all-windows baseline succeeds; the discrete layer is currently too sparse and too dependent on the steering-law objective. Eleventh, deterministic dense-schedule repair can make the phase-shift benchmark refinable for all samplers, but only by collapsing them to the same all-windows refined envelope, so the repair is diagnostic rather than

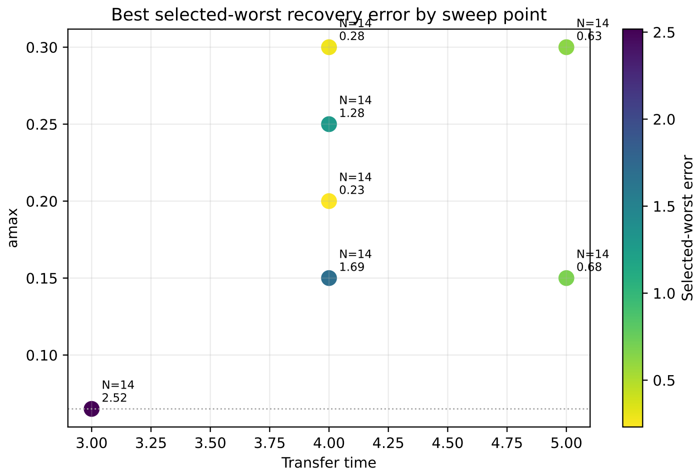

Figure 12: Feasibility sweep heatmap for the catalog-derived benchmark. The completed cases remain outside the configured robust success thresholds.

discriminating. Twelfth, adding a duty-cycle prior creates a positive but limited non-teacher result: the QUBO-ready objective can select one-coast schedules that remain robustly refinable, although exhaustive cardinality and all-windows fuel-regularization ablations show that this is a broad feasible schedule class rather than a superior Pareto frontier. Thirteenth, the 30-seed main-method statistics confirm that several high-duty samplers are reliably feasible, but all-windows continuous refinement has lower paired selected-worst error than every sampled method, and the configured statevector QAOA row does not match the strongest classical methods. Fourteenth, the 30-seed QAOA-depth statistics show that optimized-angle $p = 2$ QAOA materially improves over random-angle $p = 1$ QAOA and is competitive with the best classical QUBO sampler, but paired tests do not support superiority over surrogate-QUBO simulated annealing. Fifteenth, the hard NRHO-like to DRO-phase catalog-derived case exposes the gap between this research scaffold and mature cislunar low-thrust design: selected outage branch errors can be made small for chosen masks at higher thrust, but nominal feasibility and broad outage-family recovery remain difficult. A locked-nominal branch-recovery diagnostic, in which the nominal control is solved once and frozen so that nominal feasibility is no longer the failing component, narrows this blocker: a bounded one-segment selected subset with at least six recovery segments meets the thresholds, while the selected one/two-segment and all-single-outage scopes in that earlier fixed-final-time diagnostic still fail. The delayed-arrival h6 portfolio diagnostic then recovers all 14 one-segment masks against a delayed target with selected/all delayed worst error 0.004040 and optimizer-converged accepted branches. The updated tail-coast fixed-final-time diagnostic resolves the configured one/two hard-catalog tail-coast scope within its locked-nominal continuous-backend setup: the combined `tail_coast_all_one_two_segment_t5_portfolio` row selects and evaluates all 27 configured one- and two-segment masks at the original target and original final time, after forcing the final five nominal controls to exact zero, and reaches nominal tail-coast error 0.022992 and selected/all fixed-time worst error 0.093606. Branches with post-outage recovery variables are optimized, while two no-recovery-variable late-tail masks are threshold-feasible direct evaluations rather than optimizer-converged branches; this is why `branch_optimizer_all_success=false` despite threshold feasibility. The separate all-one-segment and all-two-segment rows remain provenance rows. This resolves the configured one/two fixed-final-time tail-coast diagnostic only within the locked-nominal continuous-backend setup; it is not fuel-optimal, not a binary-sampler result, not QUBO/QAOA/quantum evidence, not production optimal control, not high-fidelity validation, and not evidence for outage families beyond the configured one- and two-segment masks.

The teacher diagnostics are particularly revealing. Even when the binary schedule that generated the teacher target is supplied, the branch-recovery optimizer does not recover a successful selected-outage solution under the current feedback initialization. The oracle continuous-control diagnostic then succeeds on every seed. This pair of results indicates that schedule structure alone is insufficient and that the present feedback control initializer can place the least-squares refinement outside the useful basin. Continuous thrust initialization, branch selection, least-squares scaling, and recovery parameterization all need improvement before the discrete initializer can be judged fairly.

The phase-shift diagnostics point to a complementary failure mode. There, the continuous all-windows baseline succeeds without teacher controls, and the bounded-projected multiple-shooting rows confirm that a consistent norm-bounded residual can solve selected-branch versions while keeping all one-segment outage diagnostics below threshold in the successful rows. The robust-margin suite strengthens this point but also narrows it: all one-segment outage branch optimization is demonstrated in the smaller

$N = 8$, $a_{\max} = 0.3$ phase-time 0.3 and 0.4 rows, while the non-continuation phase-time 0.2 all-single-outage row fails. The continuation-extension suite then broadens the all-selected evidence: all-single-outage $N = 8$ rows at phase times 0.3 and 0.4 meet thresholds with optimizer convergence, and one/two-segment rows at phase time 0.3 meet thresholds for both $N = 6$ and $N = 8$. The same extension still fails for the $N = 6$ one/two-segment phase-time 0.2 row. The trapezoidal direct-collocation baseline independently solves the phase-time 0.4 row but fails at 0.2 and 0.3. The constant-control Hermite–Simpson continuation probe also supports phase-time 0.4 feasibility and a lower-thrust phase-shift stress row, but it preserves the same caution because the phase-time 0.2 warm row fails the nominal threshold and the feasible phase-time 0.3 cold row is not optimizer-converged. The independent-midpoint-control Hermite–Simpson package strengthens this comparison by removing the constant-control midpoint limitation and reproducing warm starts from endpoint-plus-midpoint sidecars; nevertheless, its phase-time 0.2 row still fails the nominal threshold and its catalog-DRO selected-outage row is a clear negative result. The continuous-backend concern is therefore narrowed further but not eliminated. The unrepaired binary samplers still do not select enough useful thrust windows. The dense-repair ablation confirms the interpretation: when sparse late-window candidates are expanded into the all-windows envelope, the same continuous backend succeeds, but the repair removes any meaningful method ranking. The cardinality-prior experiment then shows one way to make the binary layer useful: explicitly encode a desired duty cycle so that the sampler preserves enough low-thrust availability while still removing one window. The 30-seed main-method package strengthens that conclusion but also limits it: cross-entropy, genetic search, true-objective simulated annealing, and surrogate-QUBO simulated annealing match all-windows continuous refinement on success, while all-windows continuous refinement is better on paired selected-worst error. The separate 30-seed optimized-QAOA ablation improves the quantum-ready sampler itself: moving from random $p = 1$ angles to optimized $p = 2$ angles raises success from 15/30 to 29/30 under the same post-QUBO candidate budget. However, the paired comparison with surrogate-QUBO simulated annealing is inconclusive, the main-method statevector QAOA row is weaker than the strongest classical methods, the exhaustive ablation shows that all one-coast schedules and many two-coast schedules are feasible, and the all-windows fuel sweep shows that continuous fuel regularization is a stronger Pareto mechanism in this case. This suggests that the present binary encoding is too coarse without additional trajectory-shaping information. A stronger formulation may need multi-level thrust availability, direction classes, learned repair policies with held-out validation, or a two-stage objective that preserves dense low-thrust arcs while optimizing fuel in continuous control space.

For Aerospace Science and Technology, the present package is best framed as a methods-and-evidence paper rather than a claim of superior quantum sampling performance. It documents a realistic hybrid architecture, non-teacher solvable control cases, compact direct-collocation comparisons, robust-margin, direct multiple-shooting continuation, constant-control and independent-midpoint-control Hermite–Simpson sensitivity studies, a negative selected-outage hard-catalog envelope, fixed-final-time and delayed-arrival hard-catalog recovery diagnostics, one duty-cycle-aware discrete-initialization result, 30-seed main-method statistics, a 30-seed optimized-QAOA improvement over random-angle QAOA that remains statistically inconclusive against the best classical QUBO sampler, and several failure modes with reproducible provenance. The practical aerospace takeaway is that mission-design researchers should first secure dense continuous-control feasibility and mature continuation/collocation baselines before interpreting binary or quantum schedule sampling. Duty-cycle priors can help locate feasible high-duty schedules, but all-windows continuous fuel regularization remains stronger in this benchmark. To support a stronger performance claim, a later version would need statistically meaningful improvement over strong classical initializers under equal total trajectory-evaluation budgets, stronger QUBO/QAOA studies beyond shallow simulated sampling, and a more mature continuous optimal-control backend with larger-scale independent-control Hermite–Simpson continuation, sequential convex programming, or comparable production-grade machinery.

8. Limitations

The main limitations are as follows. The phase-shift benchmarks are non-teacher and not do-nothing passes, but they are still short normalized CR3BP retargeting cases and are easier than the NRHO-like to DRO-phase transfer. The bounded-projected multiple-shooting, direct-collocation, selected-outage hard-catalog envelope, continuation-extension, Hermite–Simpson continuation, locked-nominal, delayed-arrival, and tail-coast recovery results are continuous-backend improvements or stress tests, not binary or quantum-sampling results, and many positive selected-branch rows optimize one selected outage branch while reporting unselected outages as diagnostics using masked nominal controls. The robust-margin suite partly reduces this concern by optimizing all one-segment outage branches in the smaller $N = 8$, $a_{\max} = 0.3$ phase-time 0.3 and 0.4 rows, but it does not include two-segment outage families, and the phase-time 0.3 threshold-feasible row reached the maximum function-evaluation limit rather than optimizer convergence. The continuation-extension suite further reduces the selected-branch concern by optimizing all configured one- and two-segment outage masks in feasible $N = 6$ and $N = 8$ phase-time 0.3 rows, but the result is still a continuous-control initialization baseline and the $N = 6$ one/two-segment phase-time 0.2 diagnostic fails the robust threshold. The selected-outage hard-catalog envelope is negative evidence: selected recovery branch errors are small for selected masks, but every row fails the nominal threshold, reaches the evaluation cap with optimizer/backend success false, and retains high all-mask diagnostics.

For the hard-catalog recovery packages, the earlier fixed-final-time locked-nominal diagnostic narrows the blocker without eliminating it: it freezes a feasible nominal control and shows that a bounded one-segment selected subset with at least six recovery segments meets the thresholds, but the one positive row is threshold-feasible rather than optimizer-converged, the all-mask column is a diagnostic rather than a robustness claim, and the selected one/two-segment and all-single-outage scopes in that package still fail. The delayed-arrival h6 portfolio row optimizes all one-segment hard-catalog masks and meets thresholds with optimizer-converged accepted branches, but it changes the arrival time, uses a feasibility-oriented residual portfolio, is not fuel-optimal, and does not by itself cover fixed-final-time recovery or two-segment outage families. The tail-coast fixed-final-time diagnostic now resolves the combined configured one/two fixed-final-time tail-coast scope within its own locked-nominal continuous-backend setup, but its nominal is a locked tail-coast design with the final five controls fixed to exact zero, two late-tail no-recovery-variable branches do not count as optimizer convergence, and deterministic fallback starts are declared branch-initialization devices rather than quantum or discrete-sampler mechanisms. These hard-catalog recovery packages remain continuous-backend evidence, not quantum evidence, not fuel-optimality evidence, not production optimal control, not high-fidelity validation, and not evidence for outage families beyond the configured one- and two-segment masks.

The direct-collocation phase-shift baseline is trapezoidal and compact, not a production continuation workflow; it succeeds on only one of the three phase-suite rows. The constant-control Hermite–Simpson continuation baseline has no independent midpoint controls; its phase-time 0.3 cold row meets thresholds but not optimizer success, its phase-time 0.2 warm row fails the nominal threshold, and its lower-thrust stress row remains a catalog halo phase-shift probe rather than the hard NRHO-like tail-coast fixed-final-time result. The independent-midpoint-control Hermite–Simpson package closes that specific transcription critique and persists endpoint-plus-midpoint sidecars, but it is still continuous-backend evidence, its phase-time 0.2 warm row fails the nominal threshold, and its bounded catalog-DRO selected-outage diagnostic remains a negative result. The earlier all-outage-selected bounded phase-suite row remains infeasible. The unrepaired $N = 8$ success comes from the all-windows continuous baseline, not from the quantum-ready schedule samplers. The dense-repair ablation succeeds by converting sparse candidates to `11111111`, so it should be read as an encoding diagnostic rather than a competitive result.

The $N = 12$ duty-cycle-prior result is positive but narrow: it demonstrates refinable high-duty schedules, not lower terminal error than all-windows, not a Pareto fuel advantage over all-windows with stronger fuel regularization, and not superiority over strong classical CEM, genetic search, or true-objective simulated annealing. The 30-seed main-method package improves statistical rigor and confirms that several high-duty samplers can be feasible, but it also shows that all-windows continuous refinement has lower paired selected-worst error than every sampled method and that the configured statevector QAOA row has lower success than the strongest classical methods. Optimized-angle QAOA is tested only in statevector simulation with $p \leq 2$ and a reduced optimizer budget of one restart and ten iterations; the separate QAOA-depth 30-seed package shows improvement over random-angle QAOA, but paired tests do not support superiority over surrogate-QUBO simulated annealing. The teacher-controlled benchmark is feasible by construction and should not be confused with an operational transfer. The hard catalog-derived benchmark remains unresolved for fuel-optimal recovery, production-grade optimal control, high-fidelity validation, outage families beyond the configured one/two tail-coast masks, and any quantum-advantage claim under the present evidence. Five $N = 8$ phase-shift seeds, three teacher-controlled seeds, and legacy 10-seed cardinality-prior summaries are still insufficient for strong statistical claims outside the focused 30-seed packages. The QUBO surrogate has mixed validation performance across benchmarks. Branch recovery is evaluated against all masks separately unless all configured masks are explicitly selected for optimization. The all-windows, multiple-shooting, compact direct-collocation, constant-control and independent-midpoint-control Hermite–Simpson, and continuation baselines are implementation baselines, not mature production trajectory optimizers. The oracle diagnostic is intentionally not a valid competing initializer. The experiments are normalized CR3BP studies without ephemeris dynamics, mass depletion, eclipse constraints, navigation uncertainty, thrust pointing limits beyond acceleration norm, or operational constraints.

9. Reproducibility

The repository contains Python source, configurations, source-state metadata, generated tables, figures, and result metadata. The catalog-derived pilot command is:

```
py -3.11 scripts\run_experiment.py --config
configs\pilot_replicated.yaml
```

The stronger catalog-derived candidate command is:

```
py -3.11 -m scripts.run_experiment --config
configs/q1_candidate.yaml
```

The feasibility sweep command is:

```
py -3.11 -m scripts.run_feasibility_sweep
```

The catalog multiple-shooting diagnostic artifacts are regenerated without new solves by:

```
py -3.11 scripts\run_catalog_collocation_feasibility.py
--resume --max-cases 0
```

The selected-outage hard-catalog envelope package command is:

```
py -3.11 scripts\run_catalog_feasibility_envelope.py
--config configs\hard_catalog_selected_outage_envelope.yaml
--resume
```

The locked-nominal hard-catalog branch-recovery diagnostic command is:

```
py -3.11 scripts\run_locked_nominal_recovery.py
--config configs\hard_catalog_locked_nominal_recovery.yaml
--resume
```

The delayed-arrival hard-catalog locked-nominal portfolio diagnostic command is:

```
py -3.11 scripts\run_delayed_locked_recovery.py
--config configs\hard_catalog_delayed_recovery.yaml
--resume
```

The fixed-final-time tail-coast hard-catalog locked-nominal recovery command is:

```
py -3.11 scripts\run_tail_coast_recovery.py
--config configs\hard_catalog_tail_coast_recovery.yaml
--resume
```

The bounded-projected multiple-shooting phase-shift diagnostic command that reproduces the positive row is:

```
py -3.11 scripts\run_bounded_collocation_benchmark.py
--resume --segments 12 --selected-outages 1
--min-recovery-segments 1 --max-nfev 40
```

The bounded-projected phase-suite command is:

```
py -3.11 scripts\run_bounded_phase_suite.py --resume
```

The robust-margin suite command is:

```
py -3.11 scripts\run_robust_margin_suite.py --resume
```

The continuation-extension suite command is:

```
py -3.11 scripts\run_continuation_margin_suite.py
--config configs\continuation_extension_suite.yaml --resume
```

The compact trapezoidal direct-collocation baseline command is:

```
py -3.11 scripts\run_direct_collocation_baseline.py
```

The constant-control Hermite–Simpson continuation baseline/probe command is:

```
py -3.11 scripts\run_hermite_simpson_continuation.py --resume
```

The independent-midpoint-control Hermite–Simpson continuation evidence command is:

```
py -3.11 scripts\run_independent_hs_continuation.py
--config configs\independent_hs_continuation_baseline.yaml
--source-states data\source_states.json --resume
```

The targeted catalog feasibility command used in the additional negative test is:

```
py -3.11 -m scripts.run_feasibility_sweep --config
configs/catalog_targeted_feasibility.yaml --transfer-times 4.0
--amax 0.3 --segments 14 --max-nfev 250 --multistart
--random-starts 3 --include-bang-bang --min-recovery-segments 4
```

The non-teacher catalog halo phase-shift benchmark command is:

```
py -3.11 scripts\run_experiment.py --config
configs\q1_phase_shift.yaml
```

The deterministic dense-schedule repair ablation for the same benchmark is:

```
py -3.11 scripts\run_experiment.py --config
configs\q1_phase_shift_repair.yaml
```

The 12-segment duty-cycle-prior phase-shift benchmark command is:

```
py -3.11 scripts\run_experiment.py --config
configs\q1_phase_shift_cardinality.yaml
```

The 30-seed main-method duty-cycle-prior package commands are:

```
py -3.11 scripts\run_experiment.py --config
configs\q1_phase_shift_cardinality_30seed.yaml
py -3.11 scripts\run_main_method_statistics.py --config
configs\q1_phase_shift_cardinality_30seed.yaml
```

The duty-cycle ablation and all-windows fuel-regularization sweep command is:

```
py -3.11 scripts\run_cardinality_ablation.py
```

The 30-seed optimized-angle QAOA depth ablation command is:

```
py -3.11 scripts\run_qaoa_depth_ablation.py
--config configs\qaoa_depth_ablation_30seed.yaml
--angle-restarts 1 --maxiter 10
```

The legacy 10-seed QAOA depth artifacts are retained in `data/results/qaoa_depth_ablation/`, `tables/qaoa_depth_ablation/`, and `figures/qaoa_depth_ablation/`, but the QAOA-depth statistical result uses the 30-seed package above.

The teacher-controlled benchmark command is:

```
py -3.11 -m scripts.run_experiment --config
configs/q1_teacher_feasible.yaml
```

The run metadata record Python version, package versions, operating system, configuration, target-state generation, true-evaluation accounting, norm-bounded refinement, and claims limitations. The bounded phase-suite, robust-margin suite, selected-outage hard-catalog envelope package, locked-nominal hard-catalog branch-recovery diagnostic, delayed-arrival hard-catalog portfolio diagnostic, tail-coast fixed-final-time recovery diagnostic, continuation-extension suite, constant-control and independent-midpoint-control Hermite–Simpson baselines, direct-collocation metadata, 30-seed main-method statistics, and 30-seed QAOA metadata also record per-row settings fingerprints, configuration hashes, source-state identifiers, or statistical procedure details so that resumed rows and paired comparisons are traceable. The locked-nominal recovery metadata in `data/results/hard_catalog_locked_nominal_recovery/locked_nominal_recovery_metadata.json` records the locked-nominal/branch-recovery semantics, threshold rule, and per-case settings fingerprints alongside the `locked_nominal_recovery.csv` rows. The delayed-arrival metadata in `data/results/hard_catalog_delayed_recovery/delayed_locked_recovery_metadata.json` records the delayed-target semantics, portfolio acceptance rule, implementation identities, and per-case branch weight variants alongside `delayed_locked_recovery.csv`. The tail-coast metadata in `data/results/hard_catalog_tail_coast_recovery/tail_coast_recovery_metadata.json` records fixed-final-time semantics, exact-zero tail-coast nominal controls, the combined all-configured one/two-segment case row, provenance all-single and all-two-segment rows, branch portfolio and fallback accounting, no-recovery-variable branch semantics, and implementation identities alongside `tail_coast_recovery.csv`. The continuation-extension and constant-control Hermite–Simpson control sidecars record persisted nominal endpoint controls and control hashes used for warm-start provenance; the independent-midpoint-control Hermite–Simpson sidecars persist endpoint and midpoint nominal controls and record endpoint, midpoint, and combined sidecar hashes. A scoped artifact manifest with SHA-256 hashes is stored in `data/results/artifact_manifest.json`; the refreshed manifest records the current scoped file set and working-tree hashes at generation time. The pinned direct-dependency set is stored in `requirements-lock.txt`. The short clean-clone verification path is the Python 3.11 pytest suite and a local LaTeX build; long experiments should use the recorded artifacts unless regenerating evidence. The test suite checks objective/QUBO finiteness, target-mode metadata, teacher control norm bounds, norm-bounded refinement controls, branch-recovery accounting, delayed-recovery portfolio accounting, tail-coast recovery fallback and no-recovery-variable accounting, direct-collocation control bounds and resume fingerprints, QAOA angle optimization, and table-generation behavior.

10. Conclusion

This paper formulates quantum-ready robust maneuver initialization for low-thrust cislunar transfers as a binary scheduling problem with missed-thrust penalties embedded in the energy model. The proposed workflow keeps the split between quantum-ready discrete search and classical continuous trajectory optimization explicit. The generated evidence now includes non-teacher catalog halo phase-shift benchmarks that bounded-projected multiple shooting, compact trapezoidal direct collocation, constant-control Hermite–Simpson continuation, independent-midpoint-control Hermite–Simpson continuation, and direct multiple-shooting continuation solve with selected-branch recovery models in limited cases. The continuation-extension suite gives four optimizer-converged feasible continuous-backend rows out of five, including an $N = 8$ all-selected one/two-segment phase-time 0.3 row, while the phase-time 0.2 one/two-segment diagnostic remains robust-infeasible. The constant-control Hermite–Simpson probe adds threshold-feasible phase-time 0.3 and 0.4 rows plus a lower-thrust phase-shift stress row, but one feasible row is not optimizer-converged, the phase-time 0.2 row fails the nominal threshold, and the stress row is not the hard NRHO-like tail-coast fixed-final-time result. The independent-midpoint-control Hermite–Simpson package addresses the specific midpoint-control transcription critique and preserves endpoint-plus-midpoint sidecar provenance, but it still leaves the phase-time 0.2 diagnostic and bounded catalog-DRO selected-outage row unresolved. The selected-outage hard-catalog envelope shows an important fixed-final-time failure mode: selected recovery errors can be small for chosen masks, but nominal errors fail, optimizer/backend success is false at the evaluation cap, and all-mask diagnostics remain high. An earlier locked-nominal fixed-final-time branch-recovery diagnostic narrows this gap by freezing a feasible nominal control and recovering a bounded one-segment, six-recovery-segment selected subset under the thresholds, while broader selected one/two-segment and all-single-outage scopes in that package still fail. A delayed-arrival h6 portfolio diagnostic recovers all 14 one-segment hard-catalog masks against a delayed target, but it is a feasibility-horizon result rather than fixed-final-time robustness or fuel optimality. The updated tail-coast fixed-final-time diagnostic then resolves the configured one/two fixed-final-time tail-coast scope within its locked-nominal continuous-backend setup: the combined row selects/evaluates all 27 configured one- and two-segment masks at the original target and final time, with recovery-variable branches optimized, two late-tail no-recovery-variable masks treated as threshold-feasible direct evaluations rather than optimizer convergence, final-five-segment nominal coast, nominal tail-coast error 0.022992, selected/all fixed-time worst error 0.093606, and explicit fallback/no-recovery-variable accounting. These continuous-backend diagnostics address selected-branch-only and mature-baseline criticisms more directly, but they do not establish fuel-optimal recovery, production trajectory optimization, high-fidelity robustness, broader outage-family robustness beyond the configured one/two masks, or quantum advantage. The same benchmark family shows that the tested unrepaired binary schedule samplers can fail to identify a refinable thrust-window structure. A deterministic dense-schedule repair makes the phase-shift case refinable for all samplers, but only by converting their sparse candidates to the same `11111111` all-windows envelope, so it is a diagnostic of encoding sparsity rather than a performance win. A duty-cycle-prior variant gives a more constructive but still limited result: a QUBO-ready active-fraction prior lets surrogate-QUBO simulated annealing and strong classical heuristics find one-coast schedules that remain robustly refinable. The 30-seed main-method statistics confirm that result for cross-entropy, genetic search, true-objective simulated annealing, and surrogate-QUBO simulated annealing, but also show that all-windows continuous refinement has lower paired selected-worst error and that the configured statevector QAOA row is not competitive with the strongest classical methods. A separate 30-seed QAOA-depth package shows that optimized-angle statevector QAOA improves over random-angle QAOA and is competitive with the best classical QUBO sampler in that narrow ablation, but paired tests do not support superiority over surrogate-QUBO simulated annealing. Exhaustive cardinality and fuel-regularization ablations show that the one-coast schedules are one part of a broad high-duty feasible region and are not Pareto-superior to all-windows continuous refinement with stronger fuel regularization. The teacher-controlled benchmark shows that selected-branch robust recovery can be achieved by several initializers, but random search currently has the highest non-oracle success rate and the oracle diagnostic shows that continuous-control initialization can dominate the outcome. The defensible conclusion is therefore that mission-design work should secure dense continuous-control feasibility and mature continuation/collocation baselines before interpreting binary or quantum schedule sampling. A stronger claim would require duty-cycle-aware binary encodings that preserve dense low-thrust arcs while adding trajectory-shaping value, stronger continuous-control initialization, stronger surrogate and QAOA generalization, and statistically significant improvement over robust classical initializers on non-constructed benchmarks under equal total trajectory-evaluation budgets.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and Code Availability

All code and generated artifacts are available in the working repository associated with this draft. The benchmark source states are derived from the

NASA/JPL Three-Body Periodic Orbits API and recorded in `data/source_states.json`.

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Table 10: Claim-to-evidence hierarchy used in this manuscript. The table separates quantum-ready sampling claims from continuous-backend diagnostics and from claims left outside the evidence boundary.

Claim layer	Supporting artifacts	Defensible interpretation	Explicit boundary
Quantum-ready schedule initialization	QUBO, QAOA, classical sampler, duty-cycle-prior, and 30-seed packages	The binary/QUBO layer can be integrated and can find refinable high-duty schedules in the duty-cycle-prior case	No quantum advantage or QAOA superiority; all-windows continuous refinement remains a strong baseline
Continuous-backend feasibility	Bounded multiple shooting, continuation, direct collocation, and Hermite–Simpson diagnostics	Several non-teacher normalized CR3BP phase-shift rows are feasible with norm-bounded continuous controls	Implementation baselines only; not high-fidelity mission design or production optimal control
Hard-catalog tail-coast recovery	Tail-coast CSV, table, and metadata rows, led by the combined configured [1,2] case	Within the locked-nominal tail-coast continuous-backend diagnostic, the combined row selects/evaluates all 27 configured one- and two-segment masks and meets fixed-final-time thresholds	Not quantum evidence, not fuel optimality, not production optimal control, not high-fidelity validation, and not broader outage-family robustness
Remaining open claims	Limitations, metadata, and negative hard-catalog envelopes	The package identifies what remains unresolved rather than treating every diagnostic as a headline result	Requires stronger continuous-control initialization, broader validation, high-fidelity dynamics, and equal-budget superiority over strong classical initializers

Table 11: Bounded-projected multiple-shooting diagnostics on the non-teacher catalog halo phase-shift target. Controls are projected to $\|u_i\| \le a_{\max}$ inside the residual and evaluation paths.

Transfer time	amax	Segments	Selected outages	Initialization	Nominal error	Selected recovery worst error	All-outage diagnostic worst error	Fuel	Max $\|\|u\|\|$	Bound violation	nfev	Runtime (s)	Meets thresholds
0.5000	0.2000	12	1	linear	0.0479	0.0619	0.0752	0.1000	0.2000	0.0000	24	18.0024	True
0.5000	0.2000	8	8	linear	0.0414	0.3378	0.3378	0.1000	0.2000	0.0000	30	73.0455	False
0.5000	0.2000	12	12	linear	0.0396	0.3904	0.3904	0.1000	0.2000	0.0000	20	141.5146	False

Table 12: Bounded-projected multiple-shooting phase-suite diagnostics on non-teacher catalog halo phase-shift targets. The selected-branch rows optimize one outage branch; the all-outage-selected row optimizes all one-segment outage branches and remains infeasible.

Case	Phase time	Transfer time	amax	Segments	Selected outages	Nominal error	Selected worst error	All-mask diagnostic worst error	Max ||u||	Bound violation	nfev	Runtime (s)	Meets thresholds
all_outage_selected_diagnostic	0.3000	0.5000	0.2000	12	12	0.0396	0.3581	0.3581	0.2000	0.0000	40	366.6997	False
selected_branch	0.2000	0.5000	0.2000	12	1	0.1939	0.2168	0.2288	0.2000	0.0000	22	12.6848	False
selected_branch	0.3000	0.5000	0.2000	12	1	0.0479	0.0619	0.0752	0.2000	0.0000	24	17.5261	True
selected_branch	0.4000	0.5000	0.2000	12	1	0.0260	0.0083	0.0274	0.2000	0.0000	40	40.3375	True

Table 13: Compact bounded projected trapezoidal direct-collocation baseline on the non-teacher catalog halo phase-shift suite. Rows optimize one selected outage branch and report all-mask diagnostics separately.

Case	Phase time	Transfer time	amax	Segments	Selected outages	Nominal error	Selected worst error	All-mask diagnostic worst error	Max ||u||	Bound violation	nfev	Runtime (s)	Meets thresholds
selected_branch	0.2000	0.5000	0.2000	12	1	0.4512	0.4794	0.4805	0.2000	0.0000	30	43.4778	False
selected_branch	0.3000	0.5000	0.2000	12	1	0.3714	0.3774	0.3969	0.2000	0.0000	30	48.8181	False
selected_branch	0.4000	0.5000	0.2000	12	1	0.0357	0.0191	0.0602	0.1483	0.0000	11	18.9927	True

Table 14: Robust-margin suite on non-teacher catalog halo phase-shift targets. The thrust-margin rows optimize one selected one-segment outage branch and report all one-segment masks diagnostically. The all-single-outage rows optimize every one-segment outage mask for that row; this does not include two-segment outage families.

Group	Phase time	Transfer time	amax	Segments	Max nfev	Selected outages	Outage masks	All selected?	Nominal error	Selected worst error	All-mask worst error	Max ||u||	Bound violation	nfev	Runtime (s)	Meets thresholds
All single outages	0.2000	0.5000	0.3000	8	60	8	8	yes	0.0472	0.2523	0.2523	0.3000	0.0000	60	137.8249	False
All single outages	0.3000	0.5000	0.3000	8	60	8	8	yes	0.0555	0.0358	0.0358	0.3000	0.0000	60	128.1735	True
All single outages	0.4000	0.5000	0.3000	8	60	8	8	yes	0.0531	0.0150	0.0150	0.3000	0.0000	57	119.3214	True
Thrust margin	0.2000	0.5000	0.2000	12	80	1	12	no	0.1939	0.2168	0.2288	0.2000	0.0000	22	11.1493	False
Thrust margin	0.2000	0.5000	0.2500	12	80	1	12	no	0.1136	0.1418	0.1543	0.2500	0.0000	27	14.0958	False
Thrust margin	0.2000	0.5000	0.3000	12	80	1	12	no	0.0538	0.0738	0.0863	0.3000	0.0000	30	16.5662	True

Table 15: Continuation-extension continuous-backend diagnostics on normalized CR3BP catalog halo phase-shift targets. This is a direct multiple-shooting continuation result, not a quantum, QUBO, QAOA, or discrete-sampler result. The all-single-outage rows select all 8 one-segment masks; the one/two-segment rows select all configured one- and two-segment masks for their segment count.

Group	Phase time	Warm start	Segments	Outage masks	Max nfev	Nominal error	Selected worst error	All-mask worst error	Optimizer success	MS success	nfev	Runtime (s)	Meets thresholds
all single headroom	0.3000	cold	8	8/8	120	0.0555	0.0351	0.0351	True	True	77	138.7051	True
all single headroom	0.4000	from 0.3	8	8/8	100	0.0531	0.0139	0.0139	True	True	61	120.8934	True
two segment budget test	0.3000	cold	6	11/11	120	0.0592	0.0690	0.0690	True	True	74	106.8293	True
two segment budget test	0.2000	from 0.3	6	11/11	160	0.0651	0.2193	0.2193	False	False	160	253.8218	False
two segment budget test	0.3000	cold	8	15/15	120	0.0612	0.0435	0.0435	True	True	76	243.8341	True

Table 16: Constant-control Hermite–Simpson continuation baseline/probe on normalized CR3BP catalog halo phase-shift targets. Warm rows reuse persisted nominal-control sidecars. This is continuous-backend evidence only; the lower-thrust stress row is not the hard NRHO-like catalog benchmark.

Case ID	Group	Phase time	Warm start	Segments	Outage masks	Max nfev	Method	Nominal error	Selected worst error	All-mask worst error	Optimizer success	DC success	nfev	Runtime (s)	Meets thresholds
hs_phase_p03_cold	Phase continuation (single outage)	0.3000	cold	8	3/8	50	hermite_simpson	0.0376	0.0287	0.0901	False	True	50	145.1586	True
hs_phase_p02_warm_from_p03	Phase continuation (single outage)	0.2000	from 0.3	8	3/8	80	hermite_simpson	0.1069	0.1208	0.1791	False	False	80	249.4395	False
hs_phase_p04_warm_from_p03	Phase continuation (single outage)	0.4000	from 0.3	8	3/8	80	hermite_simpson	0.0179	0.0128	0.0628	True	True	7	25.1659	True
hs_hard_p04_amax02_warm_from_p03	Hard catalog stress probe	0.4000	from 0.3	8	3/8	60	hermite_simpson	0.0183	0.0176	0.0439	True	True	24	74.8323	True

Table 17: Independent-midpoint-control Hermite–Simpson continuation evidence. Warm rows reuse persisted endpoint and midpoint nominal-control sidecars when available. This is continuous-backend evidence only; the catalog-DRO row is a bounded selected-outage negative diagnostic.

Case ID	Group	Phase time	Warm start	Segments	Outage masks	Max nfev	Method	Nominal error	Selected worst error	All-mask worst error	Optimizer success	DC success	nfev	Runtime (s)	Meets thresholds
ihs_phase_p03_cold	Phase continuation (single outage)	0.3000	cold	8	3/8	60	hermite_simpson_midpoint	0.0358	0.0300	0.0849	False	True	60	221.3432	True
ihs_phase_p02_warm_from_p03	Phase continuation (single outage)	0.2000	from 0.3	8	3/8	90	hermite_simpson_midpoint	0.0976	0.0942	0.1694	False	False	90	419.5509	False
ihs_phase_p04_warm_from_p03	Phase continuation (single outage)	0.4000	from 0.3	8	3/8	90	hermite_simpson_midpoint	0.0194	0.0152	0.0605	True	True	7	34.3690	True
ihs_phase_p04_amax02_warm_from_p03	tighter thrust phase shift	0.4000	from 0.3	8	3/8	80	hermite_simpson_midpoint	0.0197	0.0188	0.0532	True	True	17	73.7907	True
ihs_catalog_dro_tf4_amax1_selected1	hard catalog selected outage	0.0000	cold	8	1/15	50	hermite_simpson_midpoint	4.2644	11.0843	11.0843	False	False	50	50.2186	False

Table 18: Catalog halo phase-shift benchmark. The target is generated by ballistic CR3BP phase propagation of the JPL source state, not by a low-thrust teacher.

Benchmark	Method	Comparison group	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	cross_entropy	method_comparison	0.2663	0.3042	0.3042	0.3750	56.0000	0.0000	56.0000	0.0000
catalog_halo_phase_shift	genetic	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	true_sa	method_comparison	0.2663	0.3042	0.3042	0.3750	52.0000	0.0000	52.0000	0.0000
catalog_halo_phase_shift	surrogate_qubo_sa	method_comparison	0.2663	0.3042	0.3042	0.3750	20.0000	32.0000	52.0000	0.0000
catalog_halo_phase_shift	qaoa_statevector	method_comparison	0.2242	0.2625	0.2650	0.5000	20.0000	32.0000	52.0000	0.0000
catalog_halo_phase_shift	all_windows_continuous	method_comparison	0.0418	0.0750	0.0887	1.0000	30.0000	0.0000	30.0000	1.0000

Table 19: Catalog halo phase-shift benchmark with deterministic dense-schedule repair. The table exposes the solver schedule, refinement candidate schedule, repaired refined schedule, and active fractions to show that all repaired methods collapse to the same all-windows refinement envelope.

Benchmark	Method	Schedule repair	Refined schedule source	Comparison group	Solver best schedule	Refine candidate schedule	Refined schedule	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Solver best active fraction	Refine candidate active fraction	Refined active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
catalog_halo_phase_shift	cross_entropy	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	56.0000	0.0000	56.0000	1.0000
catalog_halo_phase_shift	genetic	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	00000111	00000111	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
catalog_halo_phase_shift	true_sa	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	52.0000	0.0000	52.0000	1.0000
catalog_halo_phase_shift	surrogate_qubo_sa	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	00000111	00000111	11111111	0.0418	0.0750	0.0887	0.3750	0.3750	1.0000	20.0000	32.0000	52.0000	1.0000
catalog_halo_phase_shift	qaoa_statevector	prefix_to_last_active	heuristic dense-availability envelope: set every window through the last active solver window to available, preserving later zeros	method_comparison	mixed	mixed	11111111	0.0418	0.0750	0.0887	0.5000	0.5000	1.0000	20.0000	32.0000	52.0000	1.0000
catalog_halo_phase_shift	all_windows_continuous	identity	original solver candidate schedule	method_comparison	11111111	11111111	11111111	0.0418	0.0750	0.0887	1.0000	1.0000	1.0000	30.0000	0.0000	30.0000	1.0000

Table 20: Catalog halo phase-shift benchmark with a 12-segment duty-cycle prior targeting one coast window. QUBO/QAOA totals include 96 shared QUBO-training evaluations and 24 post-QUBO true candidate evaluations.

Benchmark	Method	Schedule repair	Refined schedule source	Comparison group	Solver best schedule	Refine candidate schedule	Refined schedule	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Solver best active fraction	Active target penalty	Refine candidate active fraction	Refined active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0954	0.1214	0.1313	0.8333	0.0833	0.8333	0.8333	120.0000	0.0000	120.0000	0.2000
catalog_halo_phase_shift	cross_entropy	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	genetic	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0675	0.0926	0.1020	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	true_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	surrogate_qubo_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	24.0000	96.0000	120.0000	1.0000
catalog_halo_phase_shift	qaoa_statevector	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0942	0.1201	0.1292	0.8333	0.0833	0.8333	0.8333	24.0000	96.0000	120.0000	0.4000
catalog_halo_phase_shift	all_windows_continuous	identity	original solver candidate schedule	method_comparison	111111111111	111111111111	111111111111	0.0386	0.0601	0.0690	1.0000	0.0833	1.0000	1.0000	19.0000	0.0000	19.0000	1.0000

Table 21: Thirty-seed $N = 12$ duty-cycle-prior phase-shift method comparison. Values summarize 210 method–seed rows. QUBO/QAOA totals include 96 shared QUBO-training evaluations and 24 post-QUBO true candidate evaluations.

Benchmark	Method	Schedule repair	Refined schedule source	Comparison group	Solver best schedule	Refine candidate schedule	Refined schedule	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Solver best active fraction	Active target penalty	Refine candidate active fraction	Refined active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
catalog_halo_phase_shift	random	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0928	0.1199	0.1277	0.8333	0.0833	0.8333	0.8333	120.0000	0.0000	120.0000	0.3667
catalog_halo_phase_shift	cross_entropy	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	genetic	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0675	0.0926	0.1020	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	true_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	120.0000	0.0000	120.0000	1.0000
catalog_halo_phase_shift	surrogate_qubo_sa	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0690	0.0941	0.1018	0.9167	0.0000	0.9167	0.9167	24.0000	96.0000	120.0000	1.0000
catalog_halo_phase_shift	qaoa_statevector	identity	original solver candidate schedule	method_comparison	mixed	mixed	mixed	0.0948	0.1208	0.1298	0.8333	0.0833	0.8333	0.8333	24.0000	96.0000	120.0000	0.4333
catalog_halo_phase_shift	all_windows_continuous	identity	original solver candidate schedule	method_comparison	111111111111	111111111111	111111111111	0.0386	0.0601	0.0690	1.0000	0.0833	1.0000	1.0000	19.0000	0.0000	19.0000	1.0000

Table 22: Thirty-seed main-method statistical diagnostics. Error deltas are method minus baseline for refined selected-worst error; negative values favor the method. Sign-test values shown as 0.0000 in the generated table should be read as rounded, not literal zero.

method	success	wilson_95_ci	delta_success_vs_all_windows_continuous	mean_error_delta_vs_all_windows_continuous	sign_p_vs_all_windows_continuous	delta_success_vs_surrogate_qubo_sa	mean_error_delta_vs_surrogate_qubo_sa	sign_p_vs_surrogate_qubo_sa
random	11/30	[0.22, 0.54]	-0.63	+0.0595 [+0.0533, +0.0659]	0.0000	-0.63	+0.0264 [+0.0201, +0.0328]	0.0001
cross_entropy	30/30	[0.89, 1.00]	+0.00	+0.0333 [+0.0327, +0.0337]	0.0000	+0.00	+0.0002 [-0.0006, +0.0009]	0.6291
genetic	30/30	[0.89, 1.00]	+0.00	+0.0325 [+0.0318, +0.0331]	0.0000	+0.00	-0.0006 [-0.0015, +0.0002]	0.3323
true_sa	30/30	[0.89, 1.00]	+0.00	+0.0334 [+0.0329, +0.0338]	0.0000	+0.00	+0.0003 [-0.0004, +0.0010]	0.8238
surrogate_qubo_sa	30/30	[0.89, 1.00]	+0.00	+0.0331 [+0.0326, +0.0335]	0.0000			
qaoa_statevector	13/30	[0.27, 0.61]	-0.57	+0.0484 [+0.0422, +0.0544]	0.0000	-0.57	+0.0153 [+0.0090, +0.0214]	0.1849
all_windows_continuous	30/30	[0.89, 1.00]				+0.00	-0.0331 [-0.0335, -0.0326]	0.0000

Table 23: QAOA depth and angle-optimization ablation on the $N = 12$ duty-cycle-prior phase-shift benchmark. All rows use 96 shared QUBO-training trajectory evaluations and 24 true post-QUBO candidate evaluations per seed.

method	runs	refinement_success_rate	refined_nominal_error_median	refined_selected_worst_error_median	refined_nominal_fuel_median	solver_true_evaluations_median	shared_qubo_training_evaluations_median	runtime_seconds_median
surrogate_qubo_sa	30	0.9333	0.0690	0.0941	0.0917	24.0000	96.0000	0.2952
exact_qubo_top24	30	0.9333	0.0690	0.0941	0.0917	24.0000	96.0000	0.2967
qaoa_random_p1	30	0.5000	0.0800	0.1055	0.0875	24.0000	96.0000	0.3222
qaoa_optimized_p1	30	0.7000	0.0682	0.0933	0.0917	24.0000	96.0000	0.3212
qaoa_optimized_p2	30	0.9667	0.0690	0.0941	0.0917	24.0000	96.0000	0.3491

Table 24: Thirty-seed QAOA/QUBO statistical diagnostics. Error deltas are method minus baseline for refined selected-worst error; negative values favor the method.

method	success	wilson_95_ci	delta_success_vs_sa	mean_error_delta_vs_sa	sign_p_vs_sa	delta_success_vs_random_qaoa	mean_error_delta_vs_random_qaoa	sign_p_vs_random_qaoa
surrogate_qubo_sa	28/30	[0.79, 0.98]						
exact_qubo_top24	28/30	[0.79, 0.98]	+0.00	+0.0006 [-0.0026, +0.0039]	0.3323			
qaoa_random_p1	15/30	[0.33, 0.67]	-0.43	+0.0119 [+0.0053, +0.0184]	0.5716			
qaoa_optimized_p1	21/30	[0.52, 0.83]	-0.23	+0.0048 [-0.0012, +0.0109]	1.0000	+0.20	-0.0072 [-0.0141, -0.0001]	0.3616
qaoa_optimized_p2	29/30	[0.83, 0.99]	+0.03	-0.0006 [-0.0039, +0.0024]	0.4545	+0.47	-0.0125 [-0.0188, -0.0060]	0.2478

Table 25: Exhaustive high-duty cardinality ablation for the $N = 12$ phase-shift benchmark.

k_active	schedules_refined	success_count	all_mask_success_count	best_nominal_error	median_nominal_error	best_selected_worst_error	median_selected_worst_error	best_all_mask_worst_error	median_all_mask_worst_error	best_nominal_fuel	median_nominal_fuel	best_schedule_by_selected_worst	min_fuel_feasible_schedule
10	66	41	41	0.0798	0.0874	0.1056	0.1132	0.1146	0.1230	0.0833	0.0833	111111111100	111111011110
11	12	12	12	0.0588	0.0617	0.0833	0.0864	0.0923	0.0956	0.0917	0.0917	111111111110	111111111011
12	1	1	1	0.0386	0.0386	0.0601	0.0601	0.0690	0.0690	0.1000	0.1000	111111111111	111111111111

Table 26: All-windows fuel-regularization sweep for the $N = 12$ phase-shift benchmark.

fuel_residual_weight	refinement_success	all_mask_success	refined_nominal_error	refined_selected_worst_error	refined_all_mask_worst_error	refined_nominal_fuel	refinement_nfev
0.0180	True	True	0.0386	0.0601	0.0690	0.1000	19
0.0300	True	True	0.0468	0.0711	0.0791	0.0962	18
0.0500	True	True	0.0773	0.1100	0.1122	0.0846	15
0.0800	False	False	0.1204	0.1491	0.1564	0.0700	10
0.1200	False	False	0.1891	0.2128	0.2183	0.0504	7
0.2000	False	False	0.2755	0.2892	0.2915	0.0268	7

Table 27: Teacher-controlled benchmark results. Values are medians across three seeds. QUBO and QAOA totals include the shared QUBO-training trajectory evaluations.

Benchmark	Method	Comparison group	Refined nominal error	Selected recovery worst error	All-mask diagnostic worst error	Active fraction	Solver true evals	Shared QUBO train evals	Total true evals	Refine success
teacher_controlled	random	method_comparison	0.0115	0.0137	1.1009	0.3750	52.0000	0.0000	52.0000	0.6667
teacher_controlled	cross_entropy	method_comparison	0.0381	0.2626	0.7695	0.3750	56.0000	0.0000	56.0000	0.3333
teacher_controlled	genetic	method_comparison	0.0213	0.5602	0.8393	0.3125	52.0000	0.0000	52.0000	0.0000
teacher_controlled	true_sa	method_comparison	0.6349	0.9638	16.4017	0.6250	52.0000	0.0000	52.0000	0.0000
teacher_controlled	surrogate_qubo_sa	method_comparison	0.0263	0.2162	0.8145	0.3750	20.0000	32.0000	52.0000	0.3333
teacher_controlled	qaoa_statevector	method_comparison	0.2495	0.2496	1.0213	0.3125	20.0000	32.0000	52.0000	0.3333
teacher_controlled	all_windows_continuous	method_comparison	6.3495	8.4578	17.8387	1.0000	3.0000	0.0000	3.0000	0.0000
teacher_controlled	teacher_seeded_schedule	method_comparison	1.2183	0.9547	1.7652	0.4375	30.0000	0.0000	30.0000	0.0000
teacher_controlled	teacher_controls_oracle_diagnostic	oracle_diagnostic	0.0067	0.0080	1.1910	0.4375	7.0000	0.0000	7.0000	1.0000

Table 28: Teacher-controlled recovery fuel and refinement effort. Fuel is reported in normalized acceleration-time units.

Method	Nominal fuel	Mean recovery fuel	Max recovery fuel	Refinement nfev	Recovery success
random	0.0579	0.0579	0.0579	24.0000	0.6667
cross_entropy	0.0892	0.0813	0.0813	30.0000	0.3333
genetic	0.0679	0.0650	0.0650	30.0000	0.0000
true_sa	0.1592	0.1355	0.1438	16.0000	0.0000
surrogate_qubo_sa	0.0733	0.0650	0.0650	27.0000	0.3333
qaoa_statevector	0.0488	0.0488	0.0488	30.0000	0.3333
all_windows_continuous	0.2595	0.2194	0.2284	3.0000	0.0000
teacher_seeded_schedule	0.1134	0.0793	0.0849	30.0000	0.0000
teacher_controls_oracle_diagnostic	0.0281	0.0272	0.0283	7.0000	1.0000

Table 29: Teacher-controlled QUBO surrogate fit diagnostics.

seed	ridge	train_rmse	val_rmse	train_r2	val_r2	val_spearman	val_pairwise_order_accuracy	val_top_k	val_top_k_recall	n_train	n_val	shared_qubo_training_evaluations	qubo_training_true_evaluations	equal_total_budget
11	0.0001	0.0001	9.4732	1.0000	0.0443	0.7143	0.7857	2	0.5000	24	8	32	32	True
23	0.0001	0.0002	13.1807	1.0000	-1.7931	0.6190	0.7857	2	1.0000	24	8	32	32	True
37	0.0001	0.0001	7.2534	1.0000	0.6466	0.5714	0.6786	2	1.0000	24	8	32	32	True